

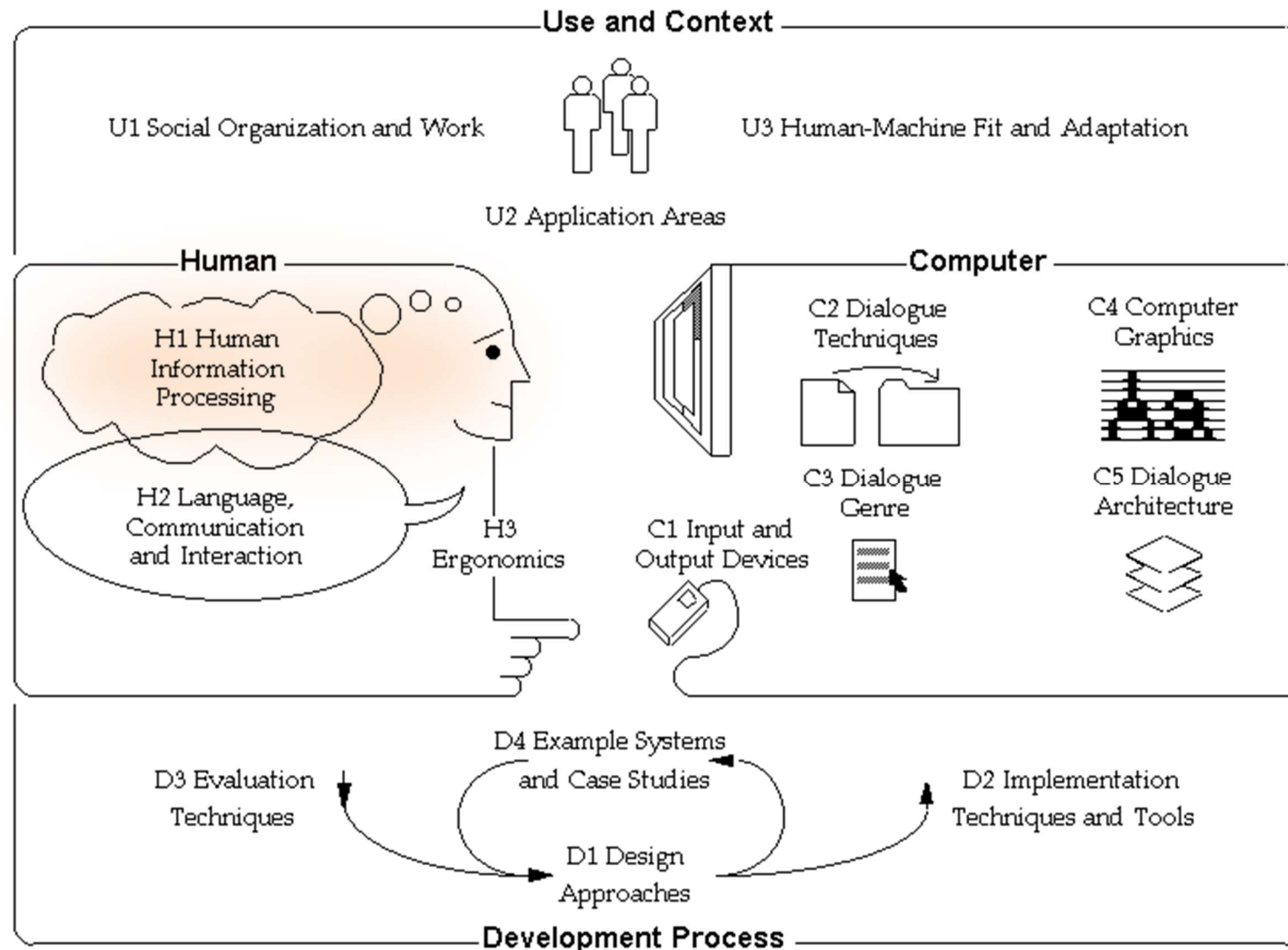
Human-Computer Interaction

Johan F. HOORN and Jeff TANG

Lecture 02: Human Information Processing, Part 1

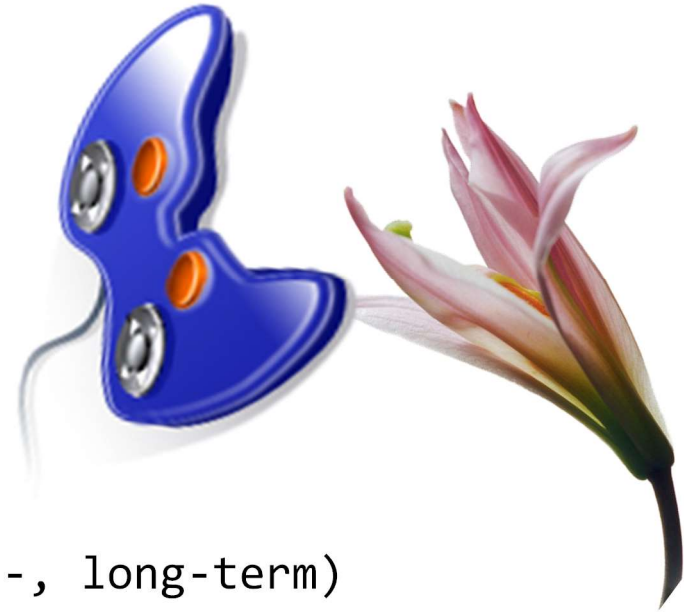


Overview



Contents

- ❑ Lecture's objectives
- ❑ The Human Information Processor
- ❑ Input: perception
- ❑ Throughput: memory (sensory, short-, long-term)
- ❑ Problem solving: reasoning, association
- ❑ Output: motor response, performance evaluation
- ❑ Summary

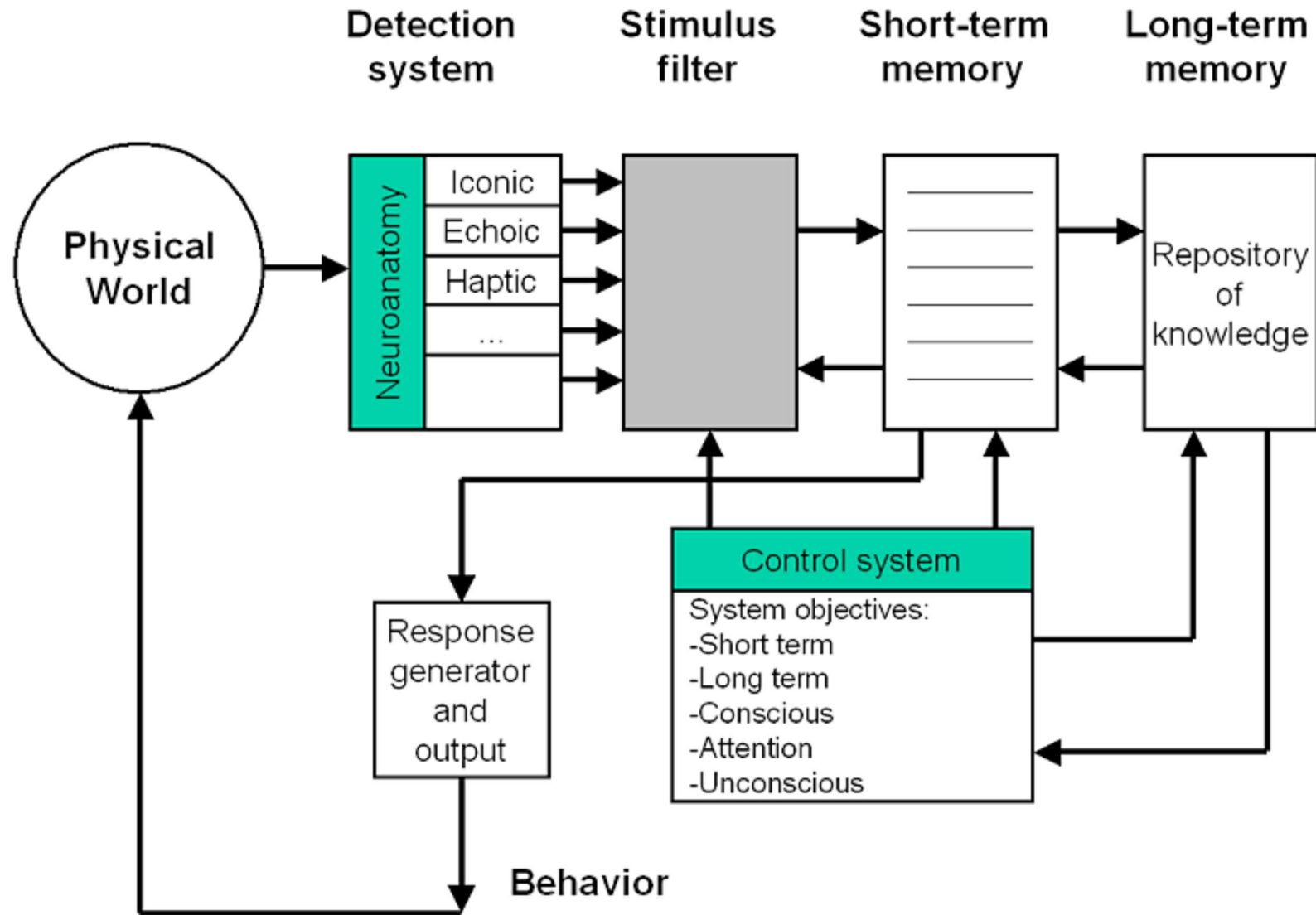


What you'll learn today

- ❑ All computer systems require human involvement
- ❑ How humans are different from computers
- ❑ Limitations and possibilities of humans
- ❑ What to watch for in system design



The Human Information Processor



Input: perception

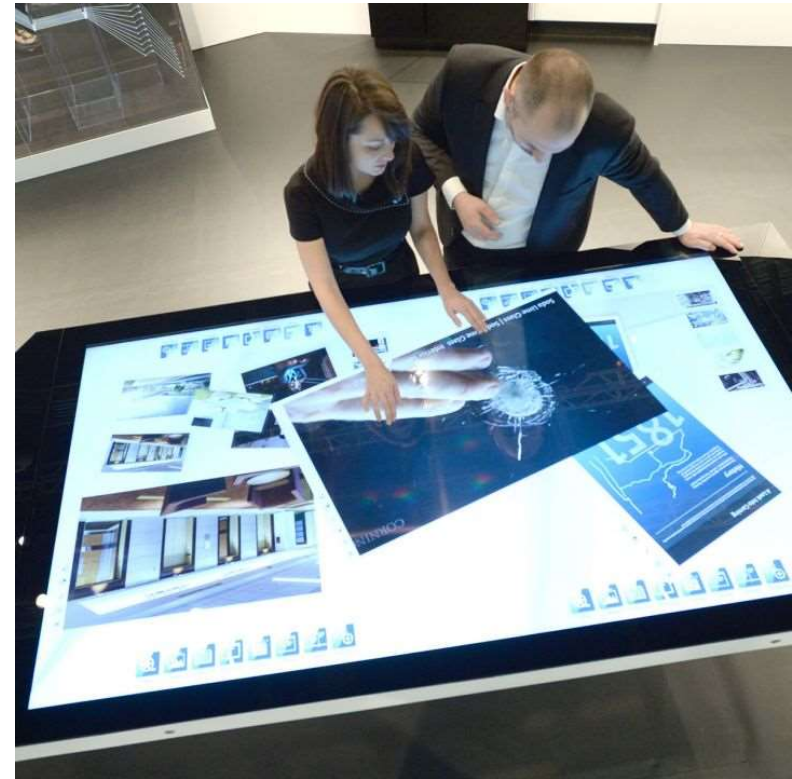
❑ Visual

- Image
 - Physical reception
 - Interpretation: 3D, color
 - Images in design
- Text
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Text in design

❑ Hearing

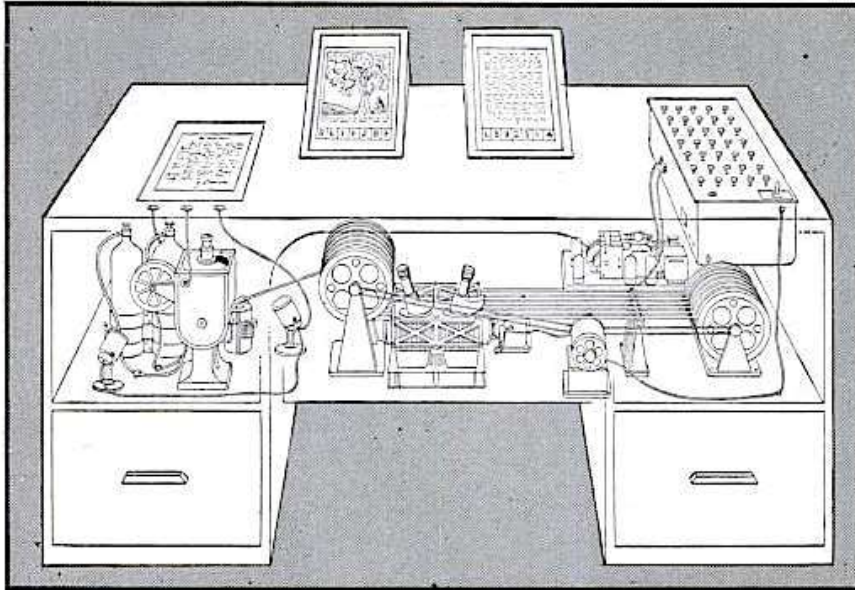
- Sound
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Sonification of the interface
- Speech
 - Physical reception
 - Interpretation: semantics, pragmatics
 - Spoken word in design

❑ Touch



Collaboration Surface Connected Home
Interactive Table | Corning

Input: perception



MEMEX in the form of a desk would instantly bring files and material on any subject to the operator's fingertips. Slanting translucent viewing screens magnify supermicrofilm filed by code numbers. At left is a mechanism which automatically photographs longhand notes, pictures and letters, then files them in the desk for future reference.

AS WE MAY THINK CONTINUED



Collaboration Surface Connected Home Interactive Table |
Corning

Vannevar Bush's Memex (Memory Extender) (1945)

Throughput: processing

❑ Memory

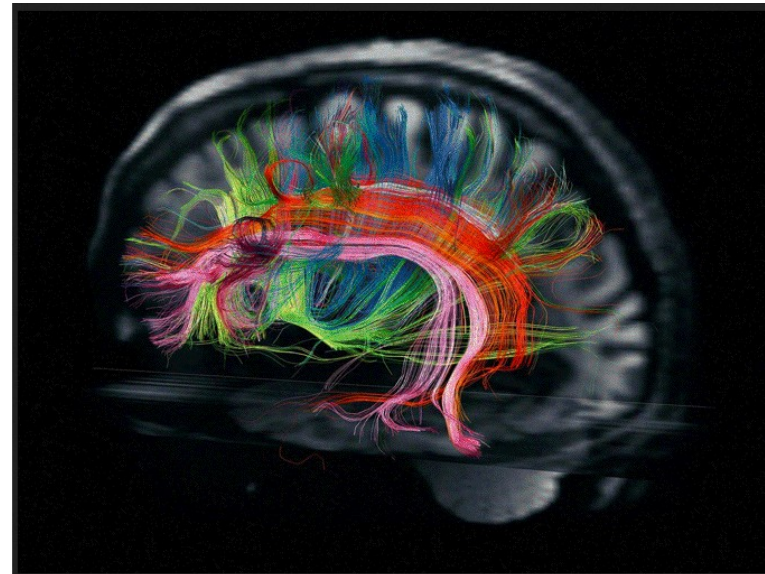
- Sensory
 - Neuroanatomy
- Short and Long Term
 - Information in the cell body

❑ Cognitive processing

- Problem solving
 - Trial and error
 - Difference reduction
 - Means-end analysis
 - Forward and backward reasoning
 - Analogical reasoning
 - Creativity

❑ Affective processing

- Goals and concerns
- Relevance
- Valence



Output: response

☐ Motor

- Speed-accuracy trade-off
 - Fitts' law
 - Shannon
- Performance evaluation
 - Index of Difficulty
 - Index of Performance
- User evaluation
 - Like?
 - Dislike?



Focus of today: Throughput

☐ Memory

- Sensory
 - Neuroanatomy
- Short and Long Term
 - Information in the cell body

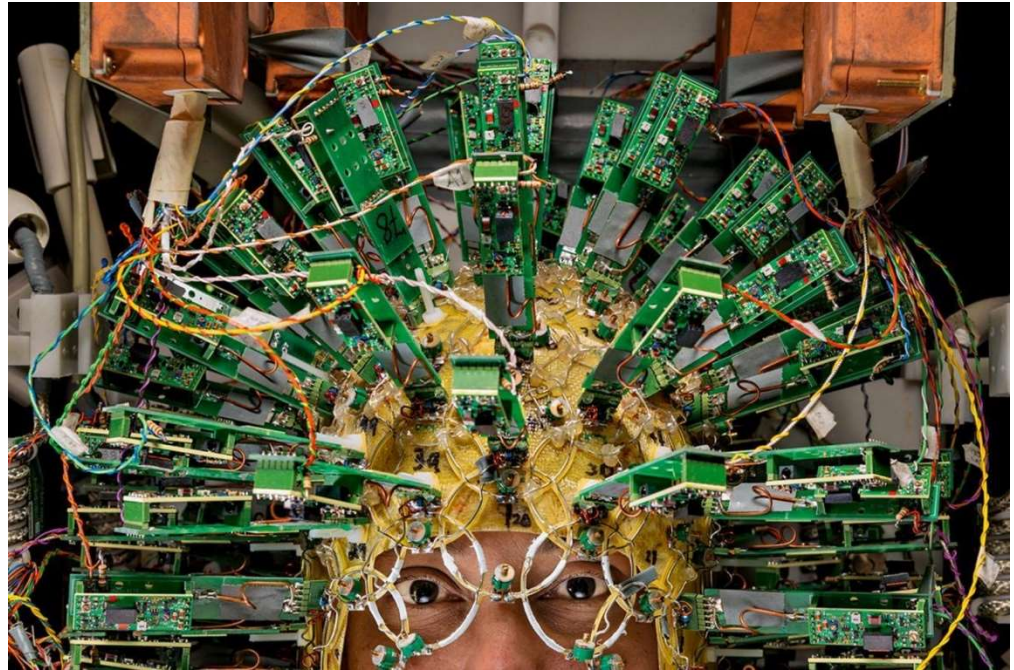
☐ Cognitive processing

☐ Affective processing

Remember Brain Computer Interfaces?

Focus of today: Throughput

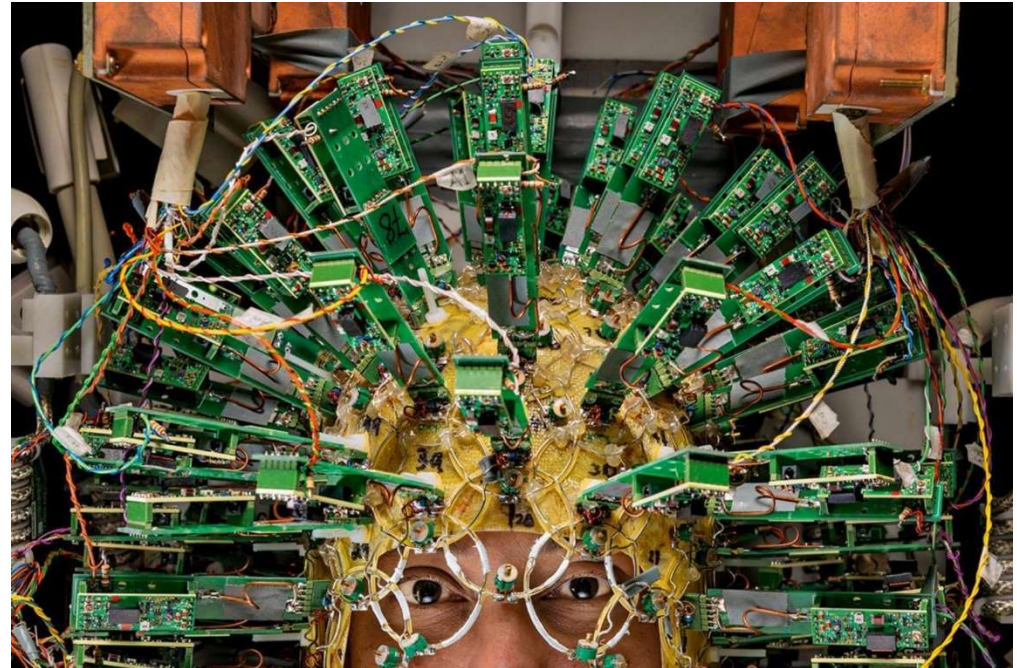
- ❑ Memory
 - Sensory
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 - Short and Long Term
 - Information in the cell body
- ❑ Cognitive processing
- ❑ Affective processing



Remember Brain Computer Interfaces?

Focus of today: Throughput

- ❑ Memory
 - Sensory
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 - Short and Long Term
 - Information in the cell body
- ❑ Cognitive processing
- ❑ Affective processing



Part of a brain scanner requiring almost as much power as a nuclear submarine. Antennas pick up signals produced when the scanner's magnetic field excites water molecules in the brain. Computers convert this data into brain maps like the one shown next (National Geographic, 2014)

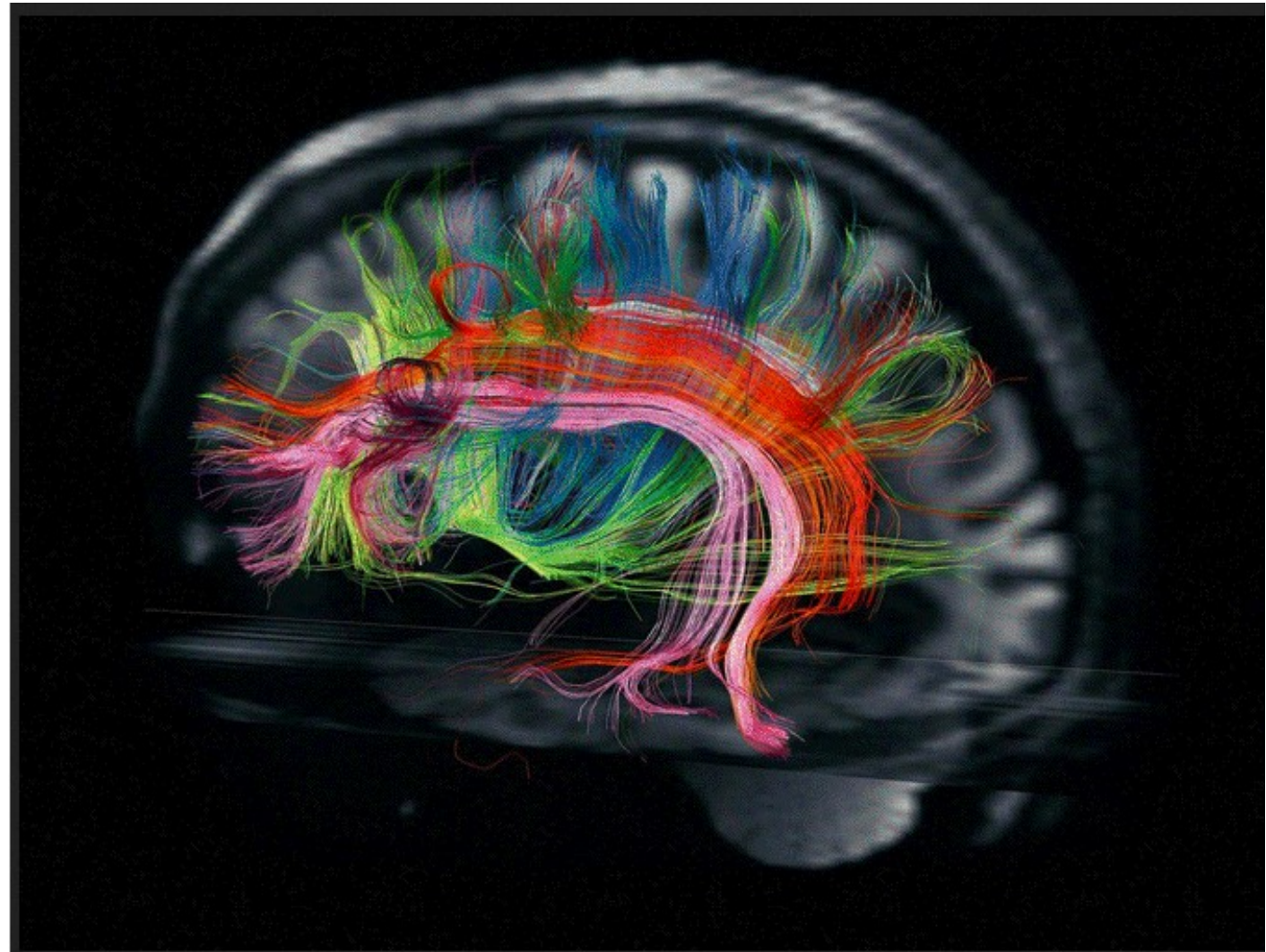
National Geographic - Secrets of the Brain. New technologies are shedding light on biology's greatest unsolved mystery: how the brain really works. By Carl Zimmer. Photographs by Robert Clark. National Geographic, February 2014. Source: <http://ngm-beta.nationalgeographic.com/2014/02/articles/secrets-of-the-brain/>

Structures and Inner Workings of the Brain

150,000 km of wiring (4 times around the earth), called 'white matter'

Weight: 1.5 kg

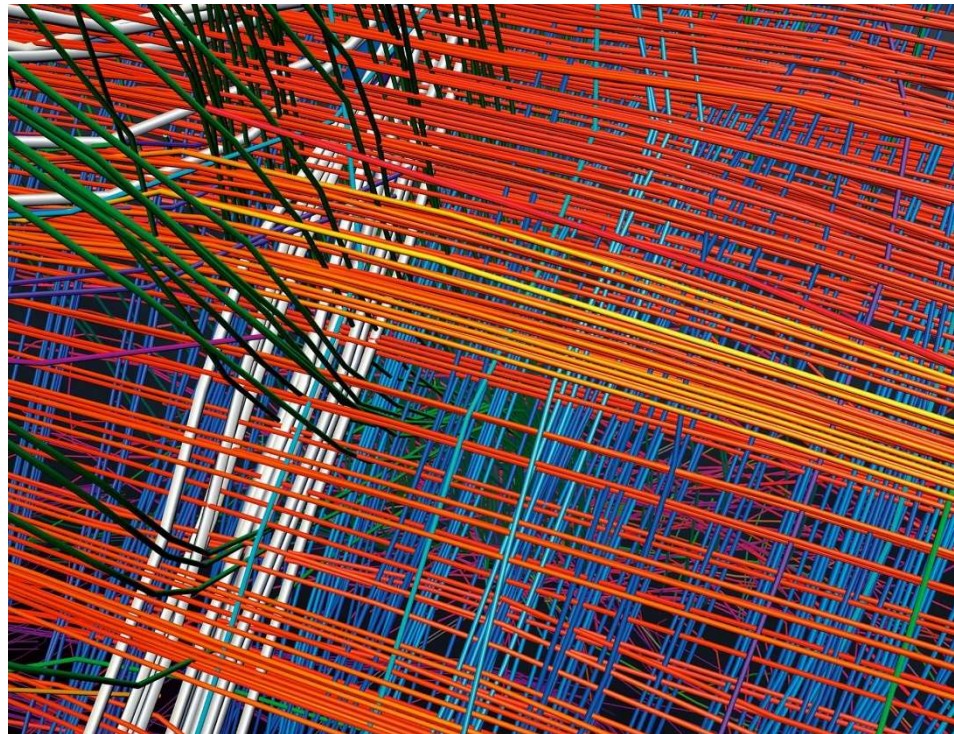
Pink and red paths (artificially colored) are involved in language processing



Structures and Inner Workings of the Brain

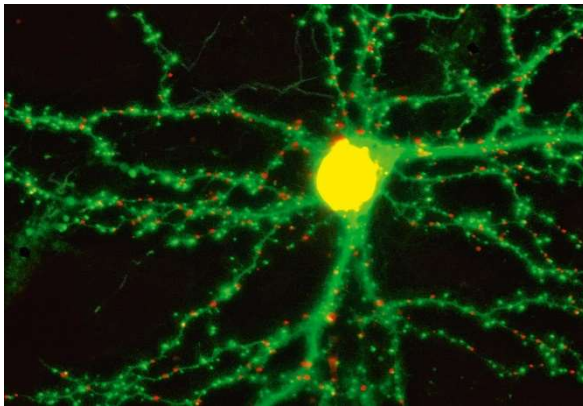
- White matter looks a bit woolly. It is not. When you zoom in, you see a frame of horizontal vs vertical lines with straight edges (Wedeen, 2012)

What we do not know yet is how information is transported over those wire frames

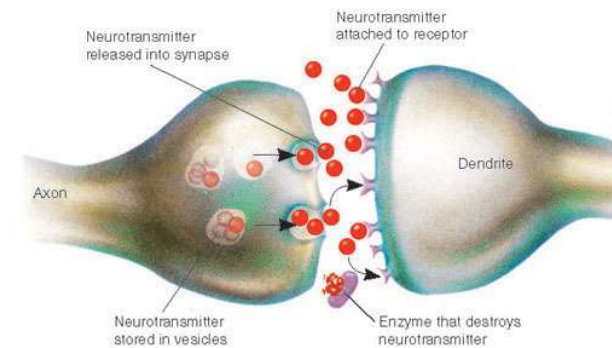
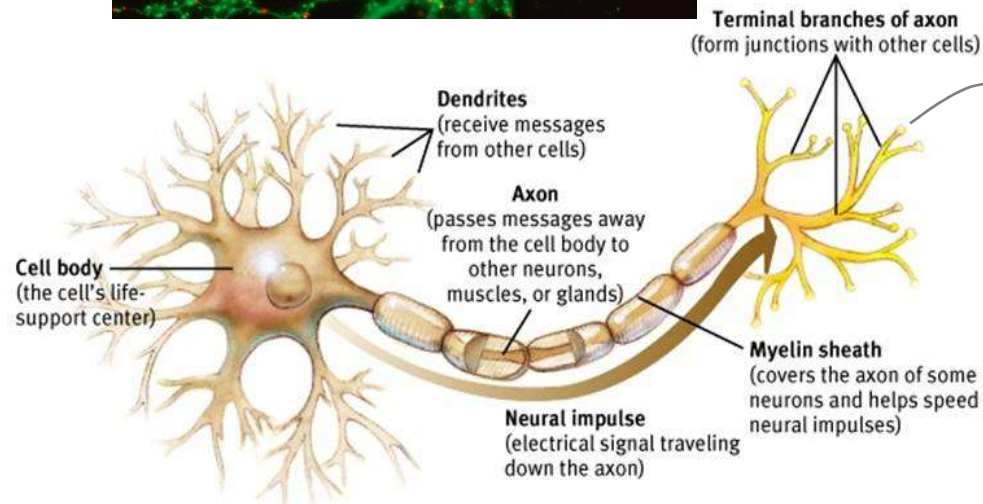


Structures and Inner Workings of the Brain

- Zooming in a little more, you see nerve cells with bumps and shoots, the 'fine structure' of the nerve cell



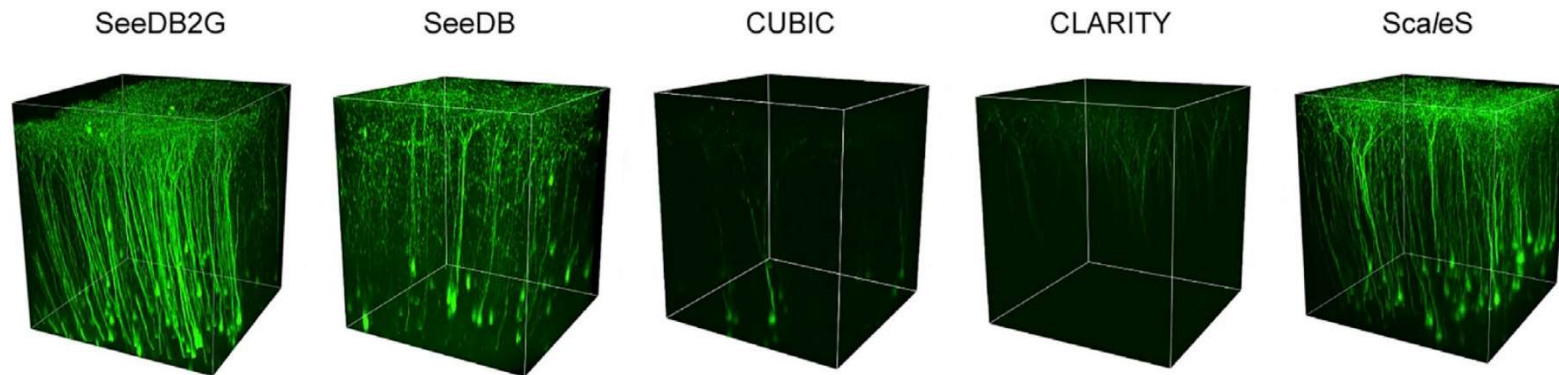
Arnold: When you remember something, things physically change in your brain. Green and red dots show contact with other cells. During the memory, new dots emerge, other fade



Synapses are electro-chemical 'switches'

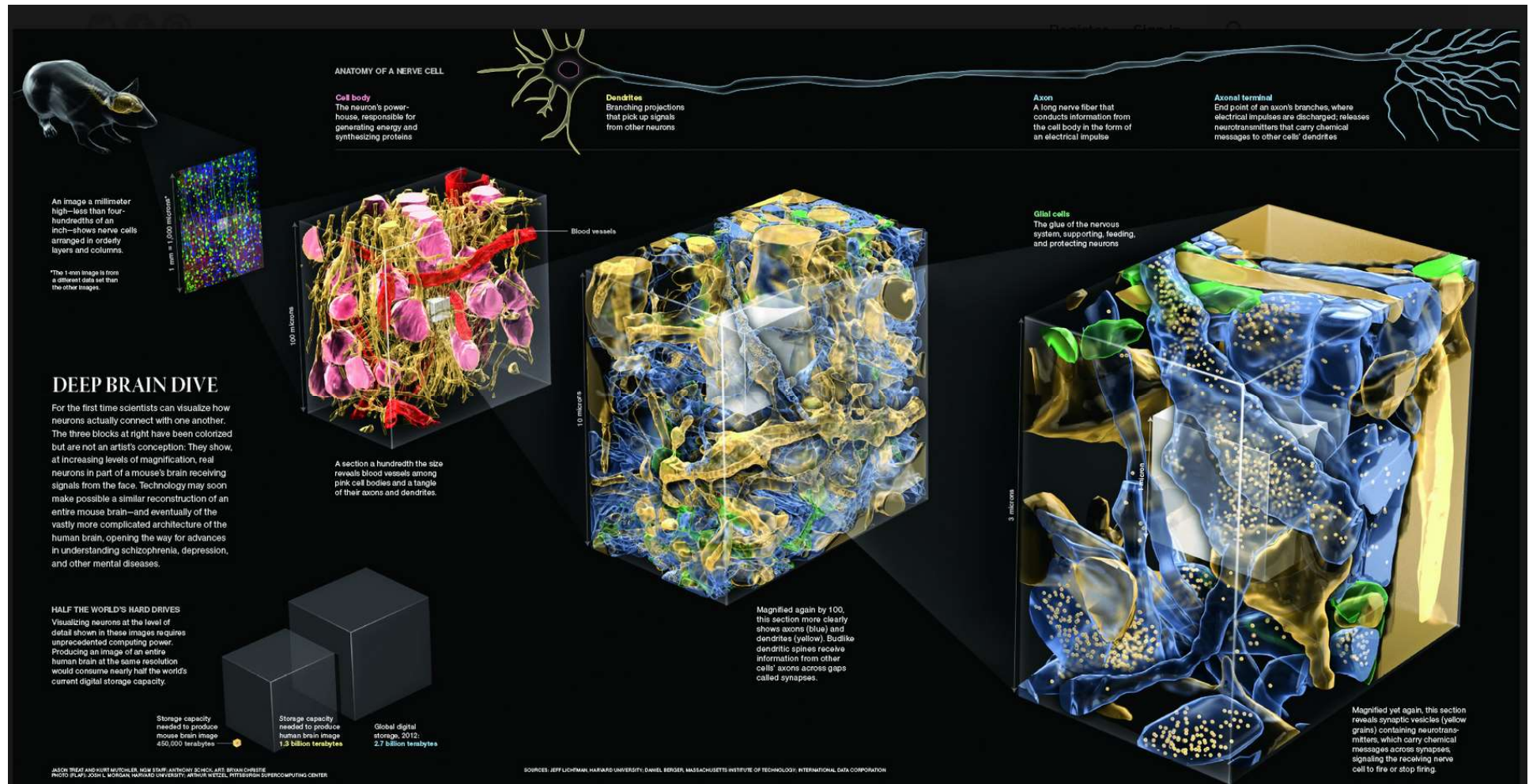
Structures and Inner Workings of the Brain

- Researchers make brain samples translucent with e.g., SeeDB2G and cut 10,000 thin slices one thousandth (.001) the size of a human hair. Then taking a picture of each with an electron microscope. Put in a row, the micro photos form a 3D model of a volume of brain tissue the size of a grain of salt

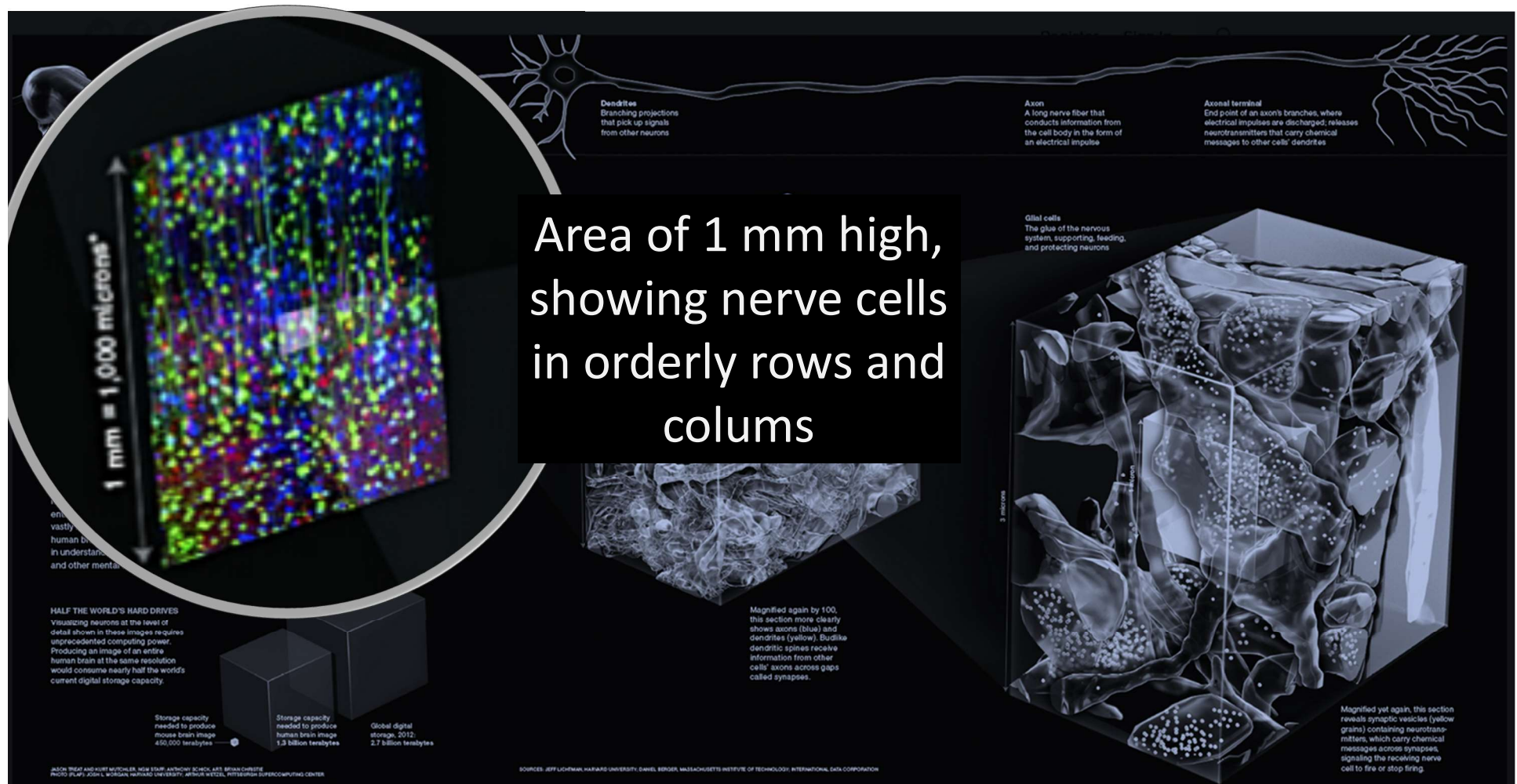


SeeDB2G provides the most detailed picture of neurons in a section of the brain

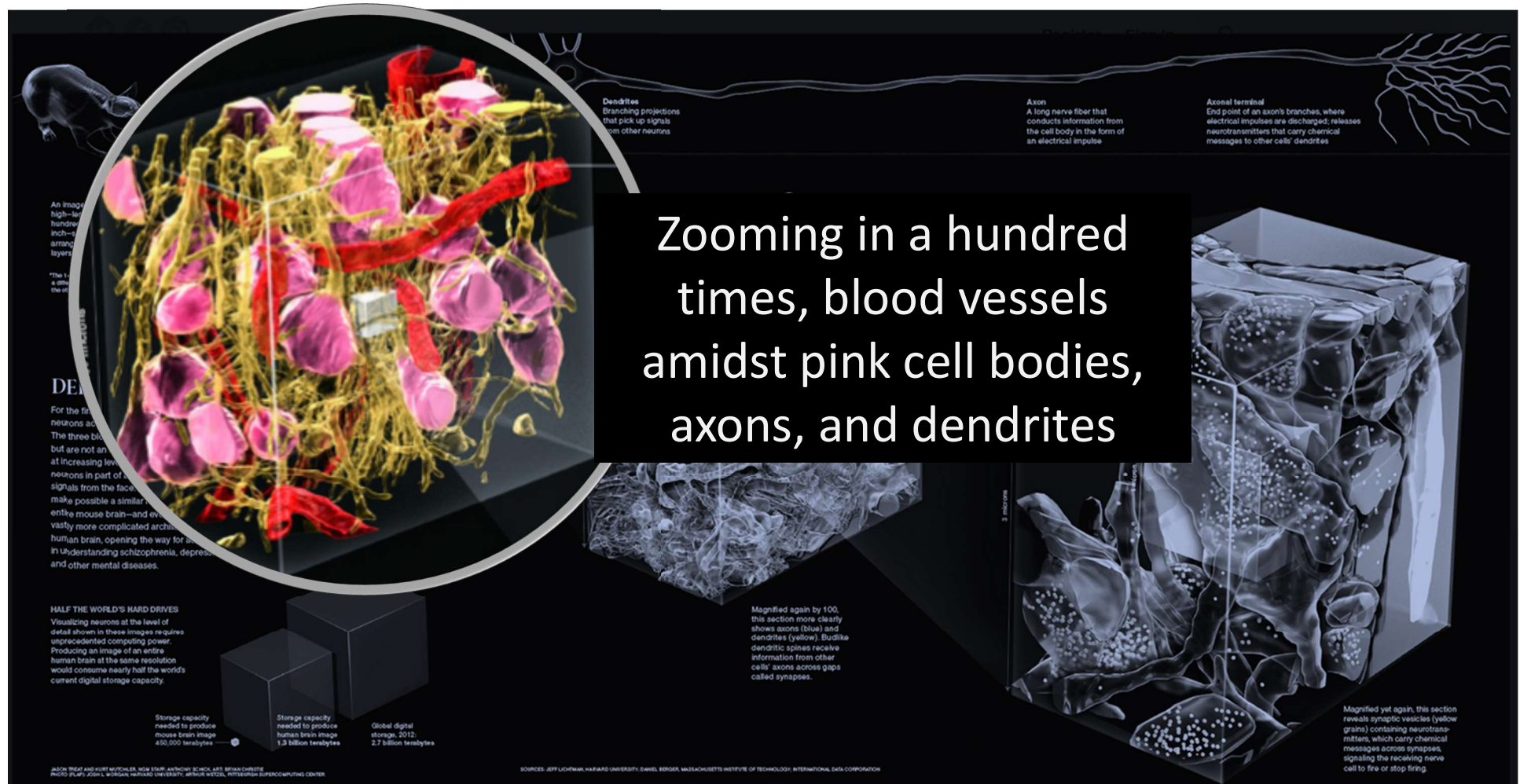
Structures and Inner Workings of the Brain



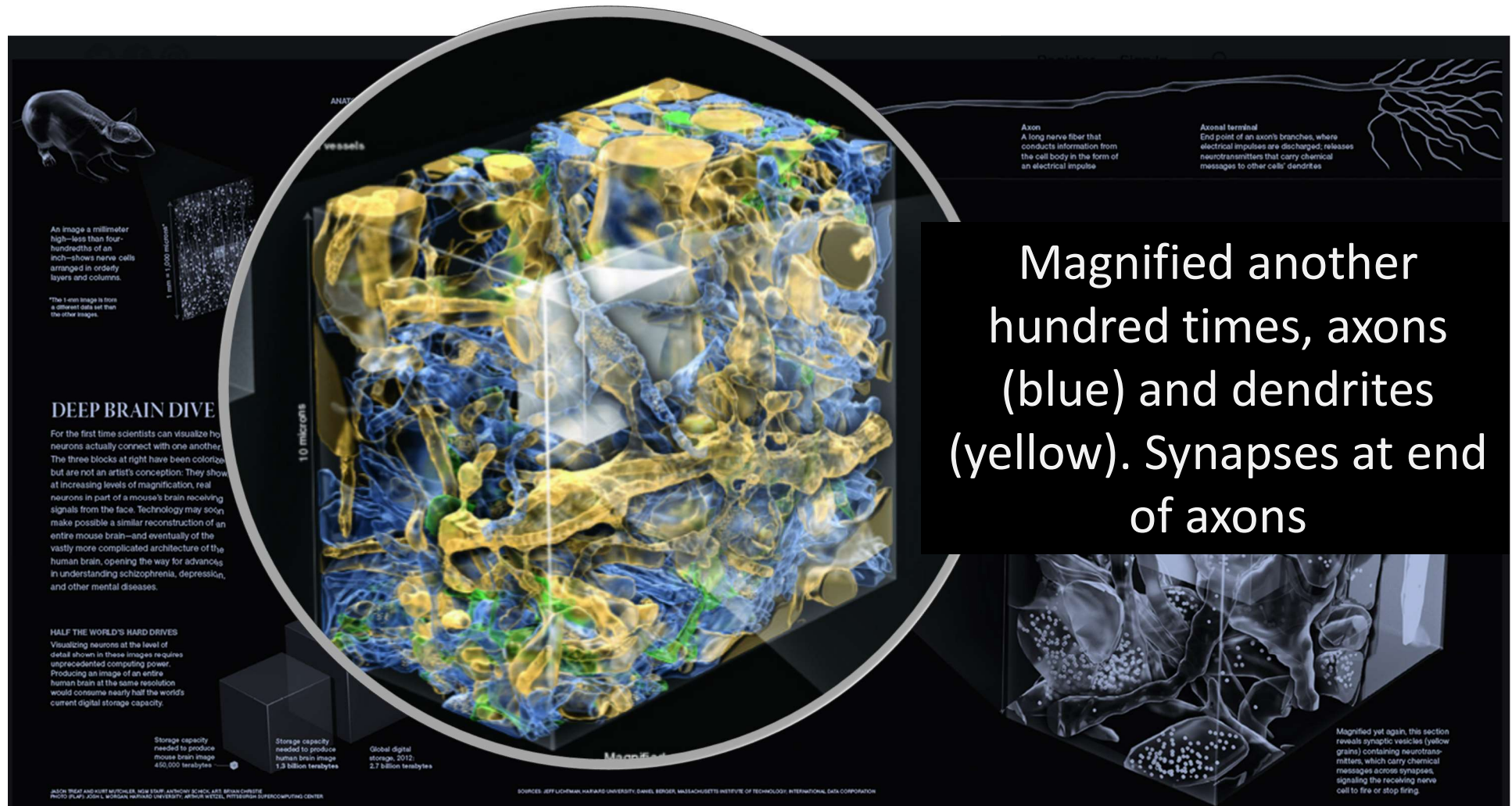
Structures and Inner Workings of the Brain



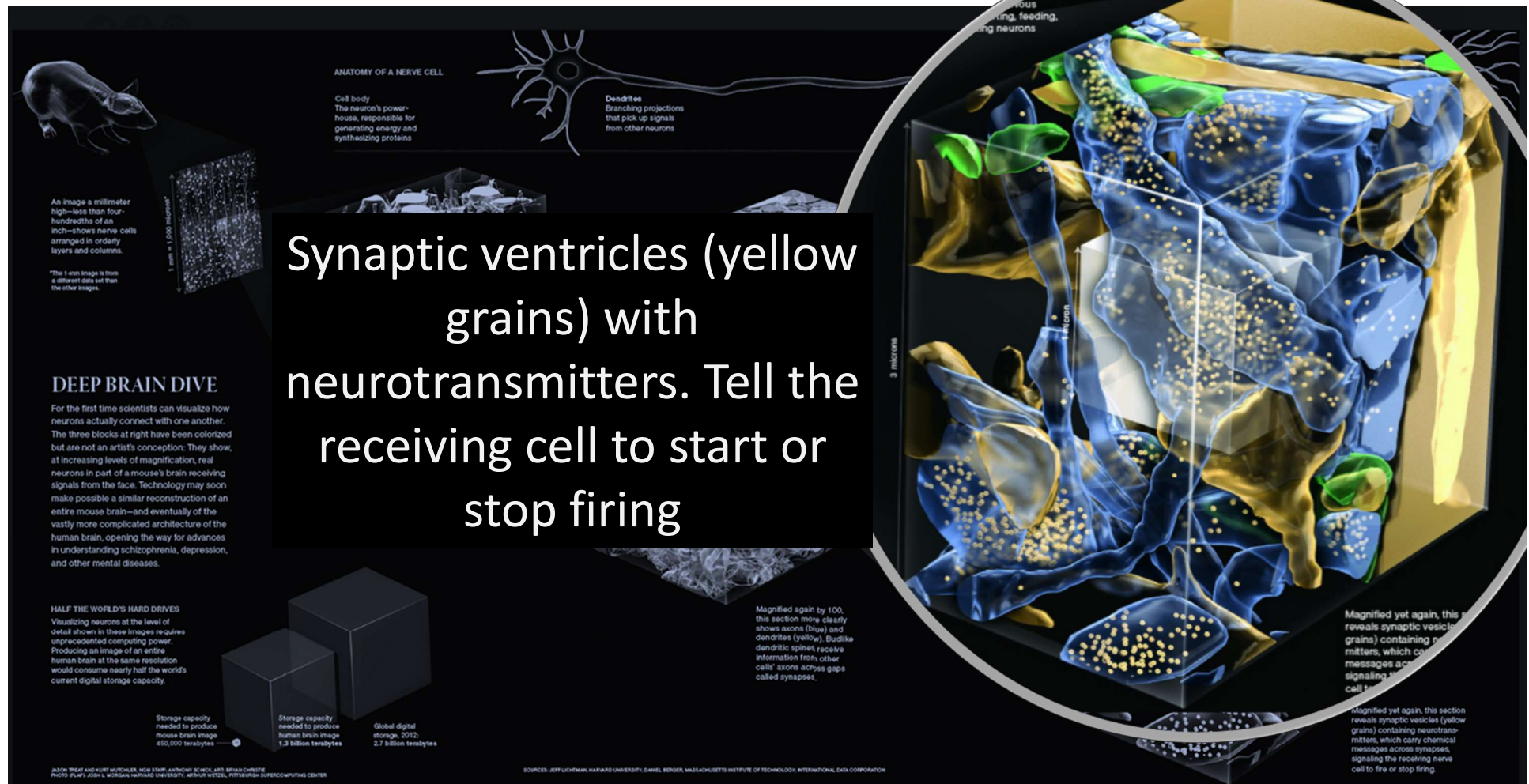
Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain



Structures and Inner Workings of the Brain

- If you take a sample of brain tissue of one thousandth (.001) micron (smaller than a grain of salt), you find about 100 axons and 80 dendrites
- Humans have about 70 billion brain cells
- That is how complex the brain is structured
- Complex, not chaotic
- Lichtman and Kasthuri discovered that each nerve cell makes contact with just one other cell, avoiding all other cells. They are very selective who to connect with

Structures and Inner Workings of the Brain

- Fried (2005) discovered the 'Jennifer Aniston neuron'
- Quian Quiroga also found brain cells for recognizing Oprah Winfrey, Golden Gate Bridge, and Eiffel Tower
- Specific information (faces, places) are stored in just one or two cells

fMRI resolution is typically around 3 x 3 x 6 mm so each sample comes from millions of neurons

Neuron 1
likes
Jennifer Aniston



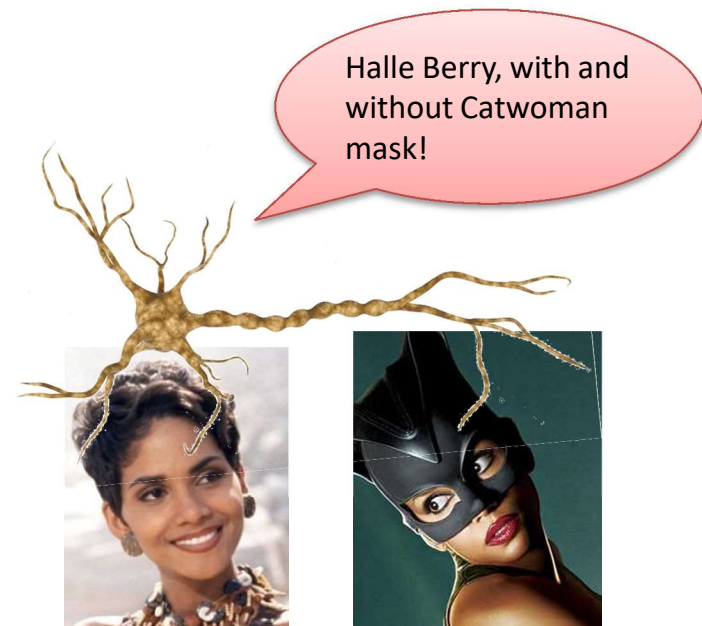
Neuron 2
likes
Julia Roberts



Neuron 3
likes
Brad Pitt



Even though there are neurons tuned to each object, the population as a whole shows no preference



Structures and Inner Workings of the Brain

Creating a False Memory in the Hippocampus

Steve Ramirez,^{1*} Xu Liu,^{1,2*} Pei-Ann Lin,¹ Junghyup Suh,¹ Michele Pignatelli,¹ Roger L. Redondo,^{1,2} Tomás J. Ryan,^{1,2} Susumu Tonegawa^{1,2†}

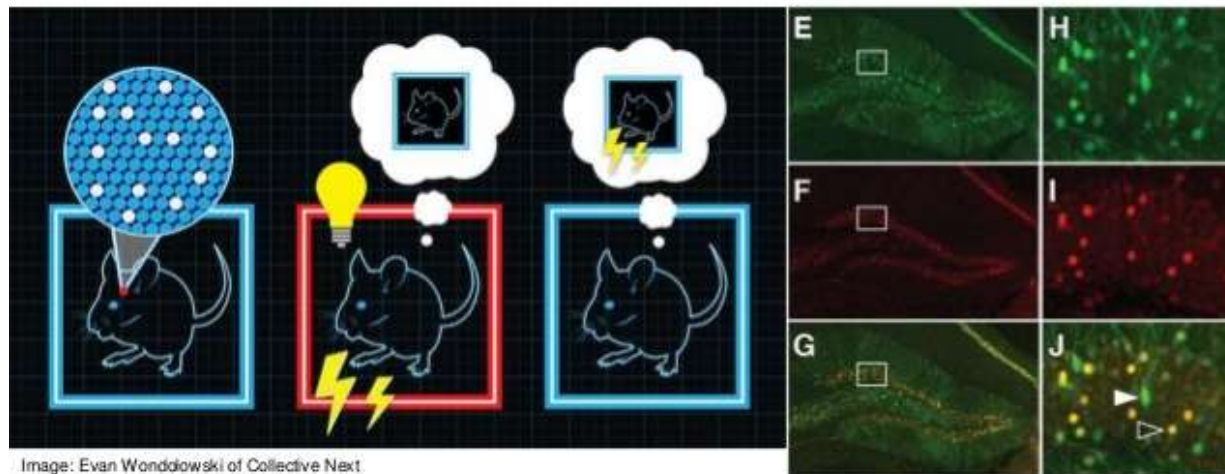


Image: Evan Wondolowski of Collective Next

Mouse in Box 1
Explores box
No fear induction

Same mouse Box 2
Shock delivered
Mouse has fear for Box 2

Same mouse Box 1
Researchers activate neurons
related to fear of Box 2
Fear response without shock

Ramirez, S., Liu, X., Lin, P. A., Suh, J., Pignatelli, M., Redondo, R. L., ... & Tonegawa, S. (2013). Creating a false memory in the hippocampus. *Science*, 341(6144), 387-391.

Structures and Inner Workings of the Brain

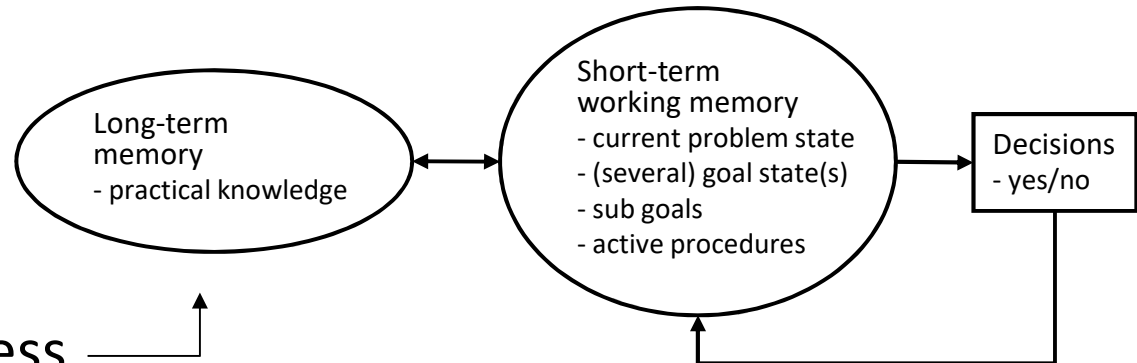
- Nerve cells are filled with DNA, proteins, and other molecules. About 84% of all genes in our DNA become active in our adult brains
- Every type of nerve cell uses a specific range of genes to build the molecular mechanism for its function
 - Light receptors in the eyes make proteins that can catch photons
 - Nerve cells in the *substantia nigra* make dopamine, crucial to our reward system (cf. affective processing)

From external reward
to internal motivation
(cf. mouse in Box 1)



What is problem solving?

- Problem
- Goal state
- Sub goals
- Problem space
- Problem-solving process



Problem solving: strategies, means, and procedures to achieve a goal

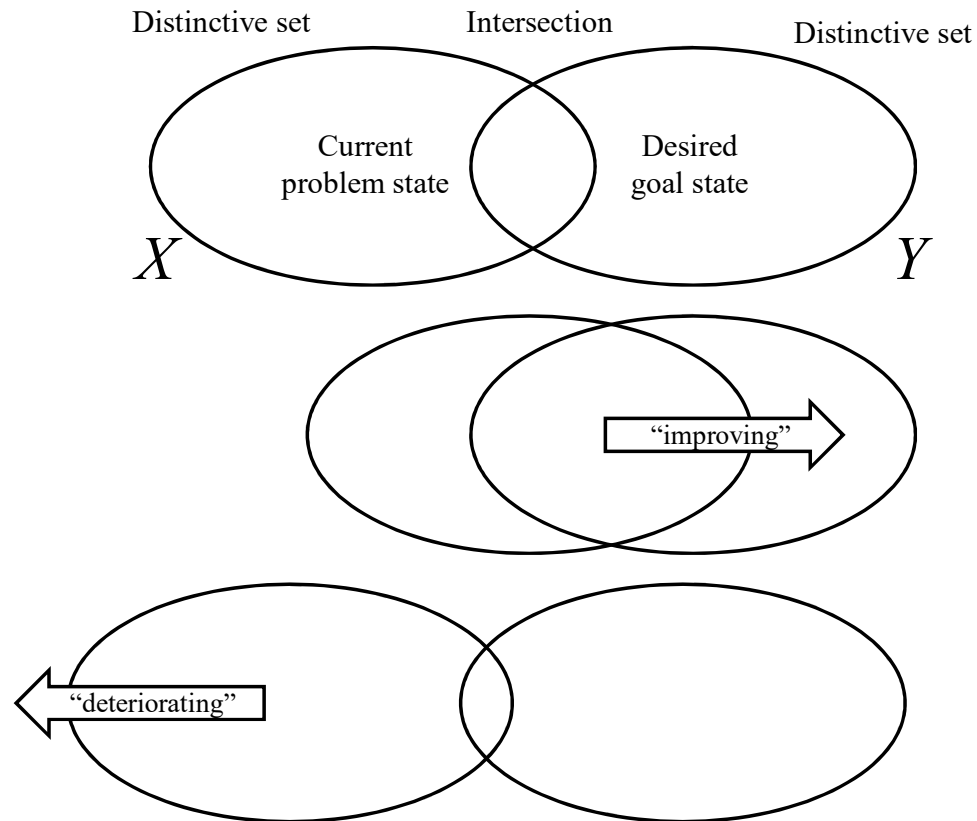
Problem: How to get what I want?

Goal state: the desired end state or solution of the problem solving activity

Sub goal: middle-term goal along the way to solving a problem

Problem space: short, middle, and long term goal states and the information needed at each step

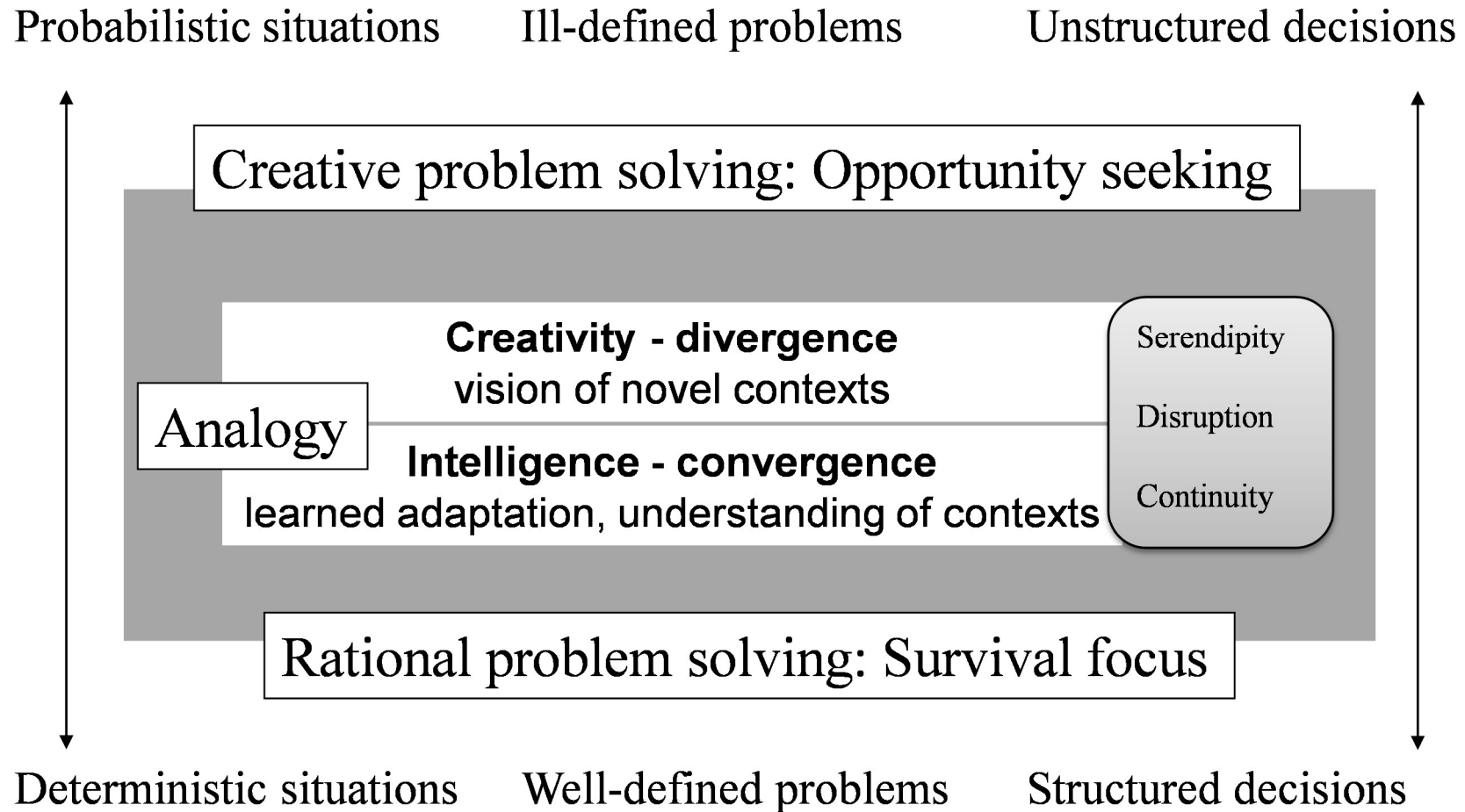
Goal states



Solving a problem: *X* should change so to overlap *Y*

Often, goal satisfaction is incomplete:
a good-enough solution

The problem continuum



The problem continuum

Well-defined problem: full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space

Example: find the bug

```
if (<boolean expression>) {  
    S1 ;  
} else {  
    S2  
}
```



```
jfhoom@flits:/home/jfhoom  
Sun Microsystems Inc. SunOS 5.8  
$ password  
bash: password: command not found  
$ passwd  
passwd: Changing password for jfhoom  
Enter login(NIS+) password: 
```

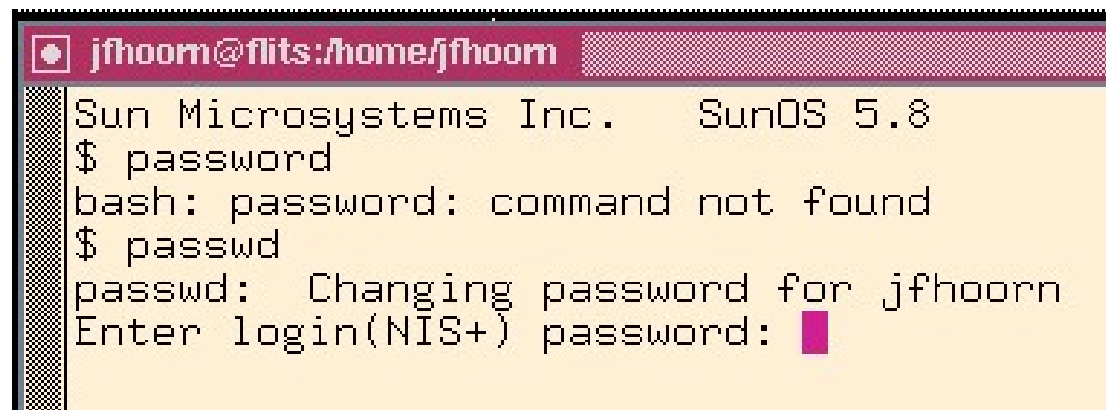
The problem continuum

Well-defined problem: full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space

Example: find the bug

```
if (<boolean expression>) {  
    S1 ;  
} else {  
    S2 ;  
}
```

missing semicolon

A terminal window with a title bar showing the user 'jfhoorn' at host 'flits' in the directory '/home/jfhoorn'. The terminal output shows the user entering '\$ password', which results in the error message 'bash: password: command not found'. The user then enters '\$ passwd', which prompts them to change their password for 'jfhoorn'. The prompt 'Enter login(NIS+) password:' is shown with a redacted password field.

```
jfhoorn@flits:/home/jfhoorn  
Sun Microsystems Inc. SunOS 5.8  
$ password  
bash: password: command not found  
$ passwd  
passwd: Changing password for jfhoorn  
Enter login(NIS+) password: 
```


ill-defined problem: partial or imprecise specification of the provided data, actions, strategies, and goal states, which cannot be (entirely) represented in a problem space

Example: mission statement

Goal decomposition:

breaking down a complex problem into smaller ones by determining sub goals

Goal state: mission



Sub goal: business goals



Sub sub goal: targets
(e.g., 5% up in two years)



Computing Services

Mission Statement

To create an online community, like no other, encompassing every facet of every municipality in the United States; to aid in online economic revitalization in partnership with local businesses and local, state and national government agencies; to publish, maintain and connect a network of community websites designed to entertain, educate and enlighten the residents of our great nation, while maintaining the highest standards of personal morals, business ethics and web etiquette.

We Provide:
Credit Commerce

**BBBONLINE[®]**
RELIABILITY PROGRAM

Problem solving strategies

Intelligence

Rule following
Difference reduction

Means-end analysis
Forward and backward reasoning

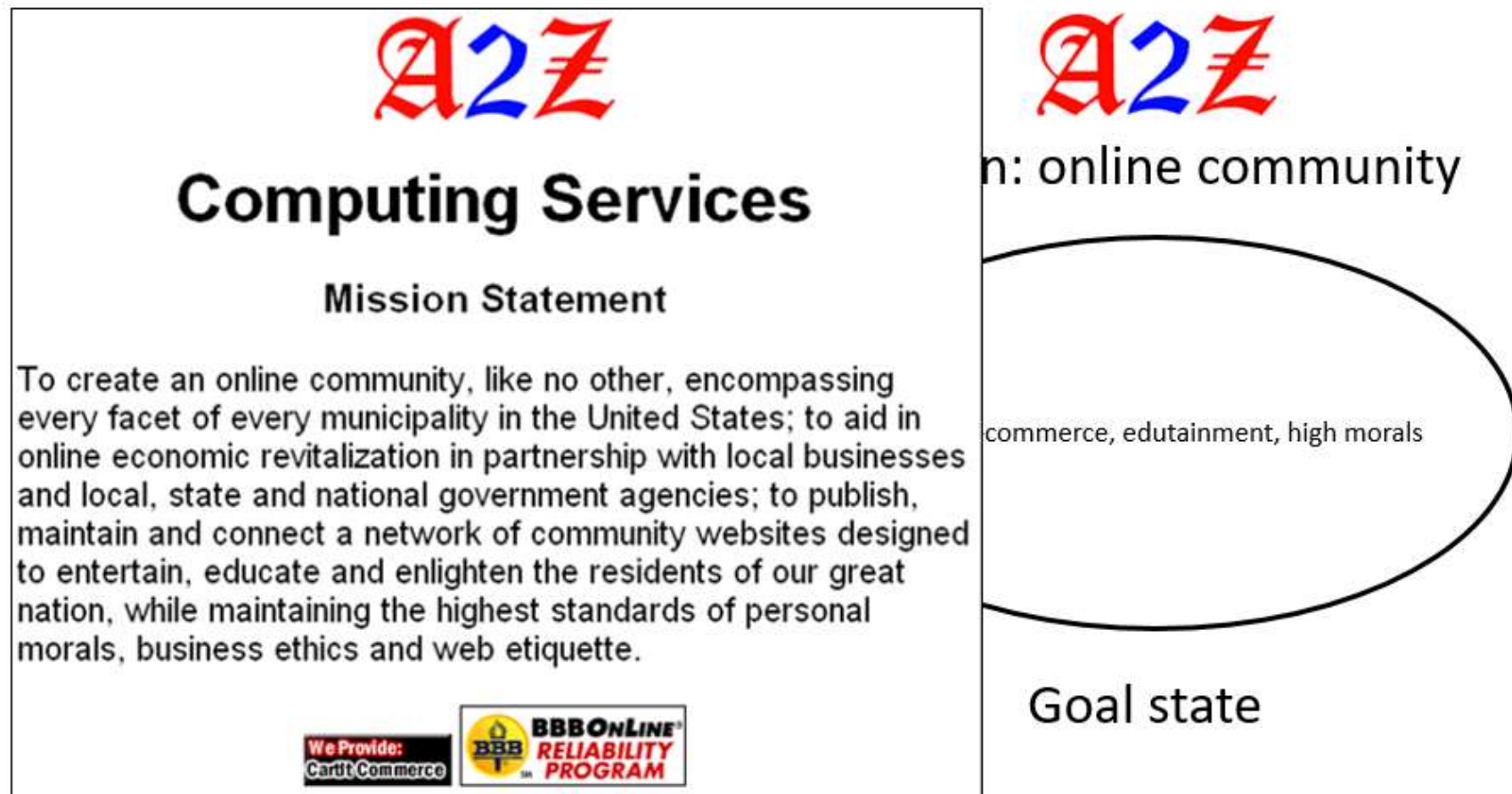
Creativity

Analogy
Random trial and error

Creative problem solving

May lead to innovation

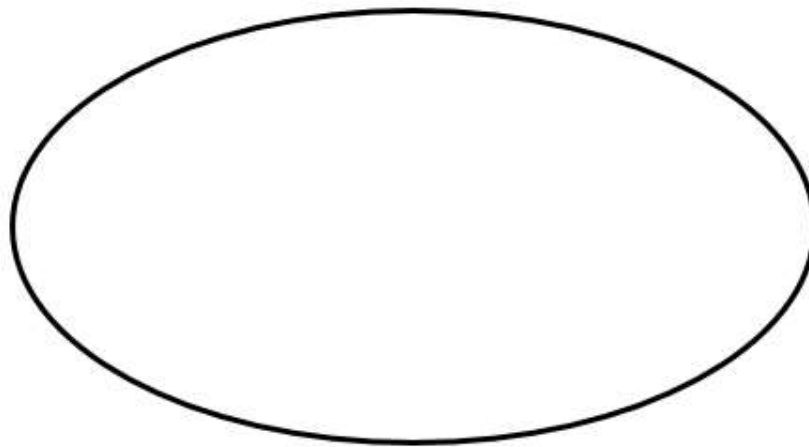
Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state



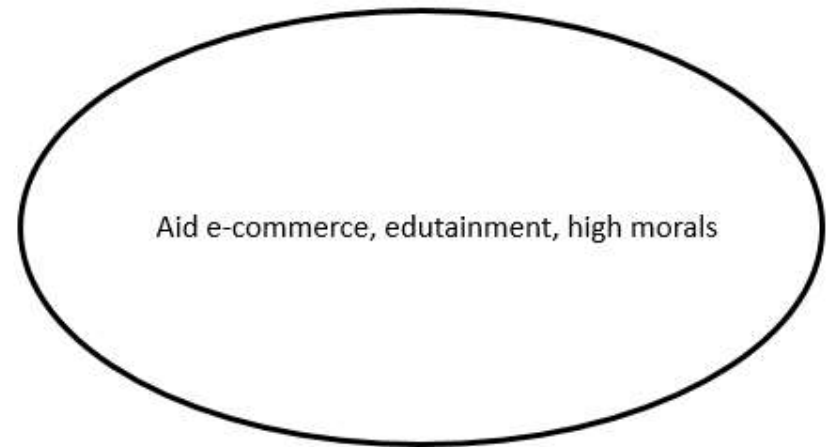
Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state

A2Z

Mission: online community



Current problem state

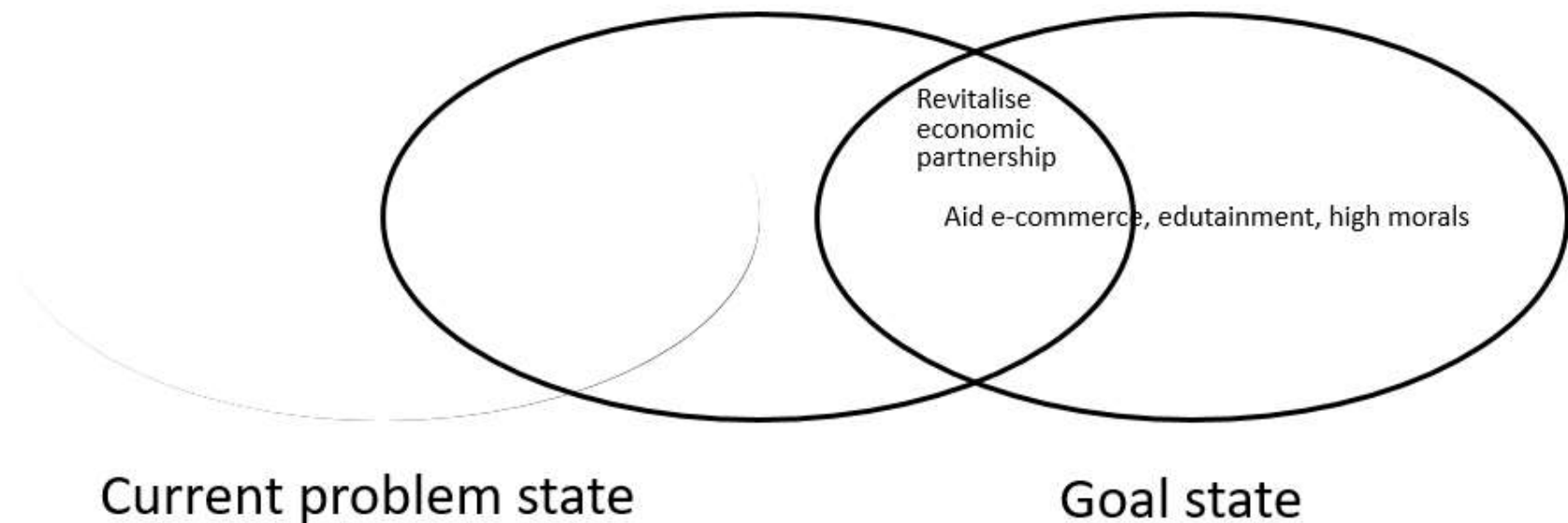


Goal state

Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state

A2Z

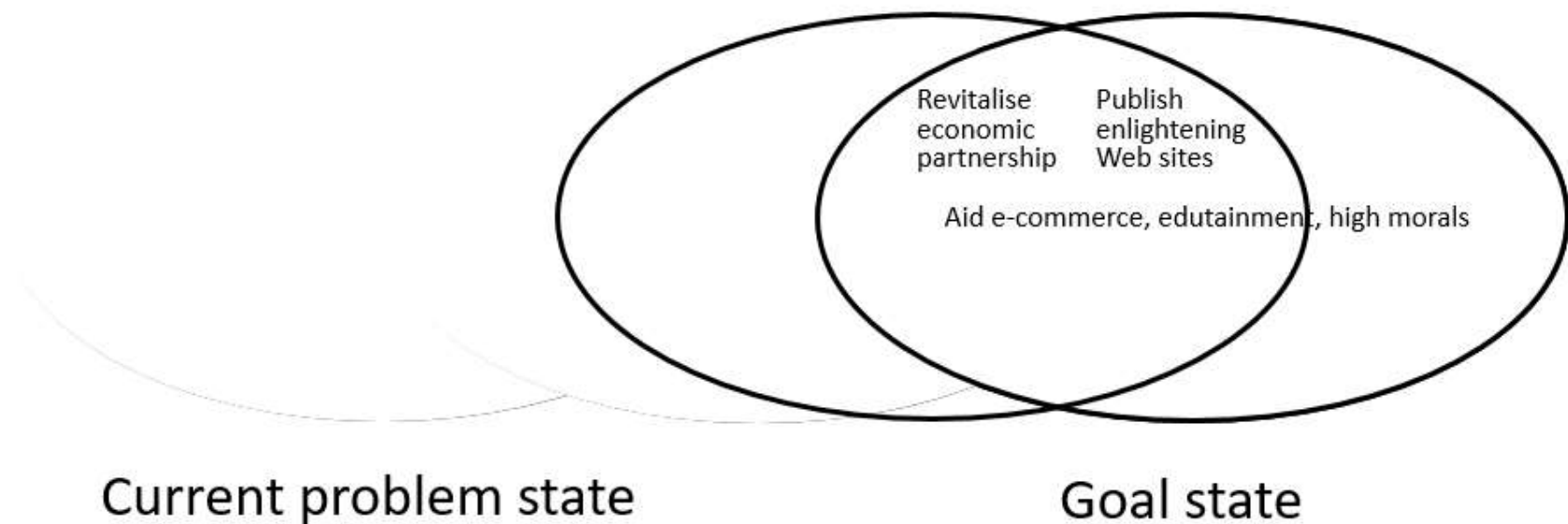
Mission: online community



Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state



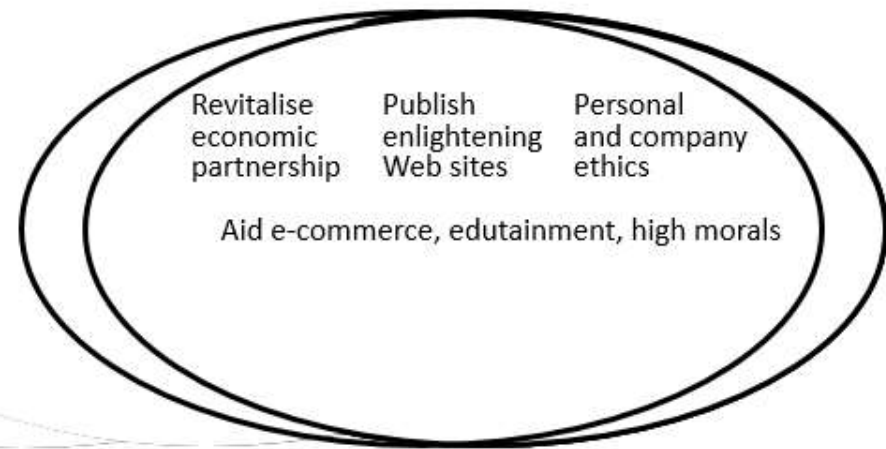
Mission: online community



Difference reduction: each sub goal that is achieved brings the current problem state closer to the goal state



Mission: online community



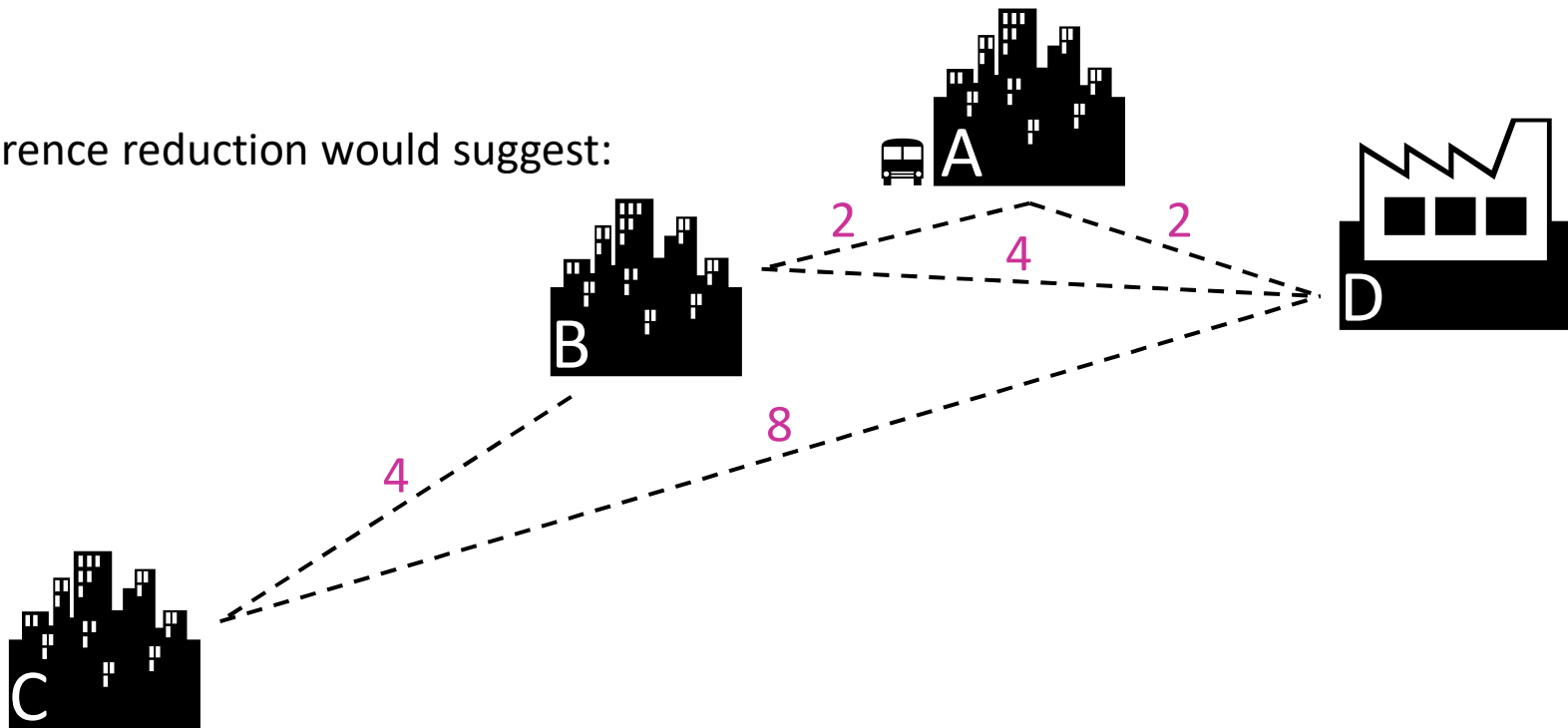
Current problem state

Goal state

Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

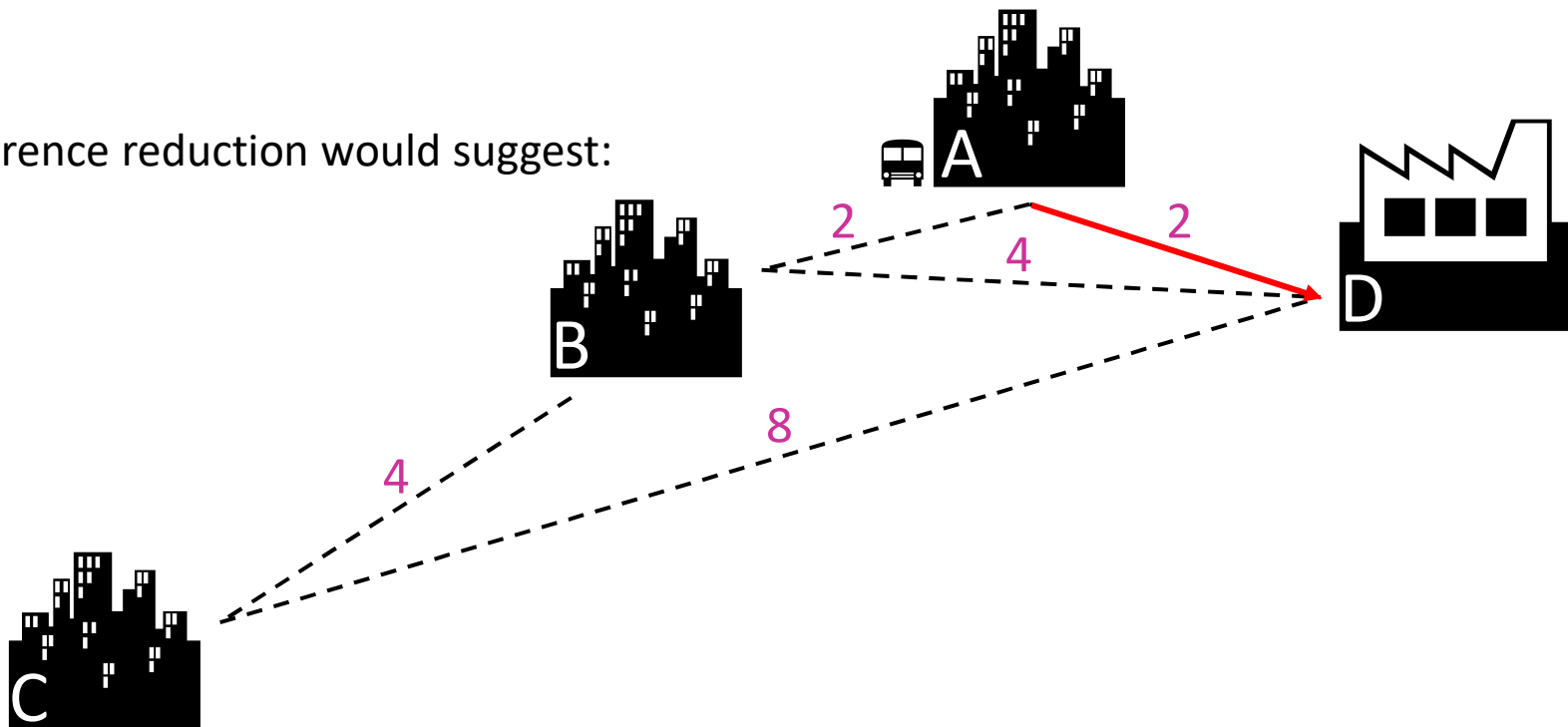


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

2 +

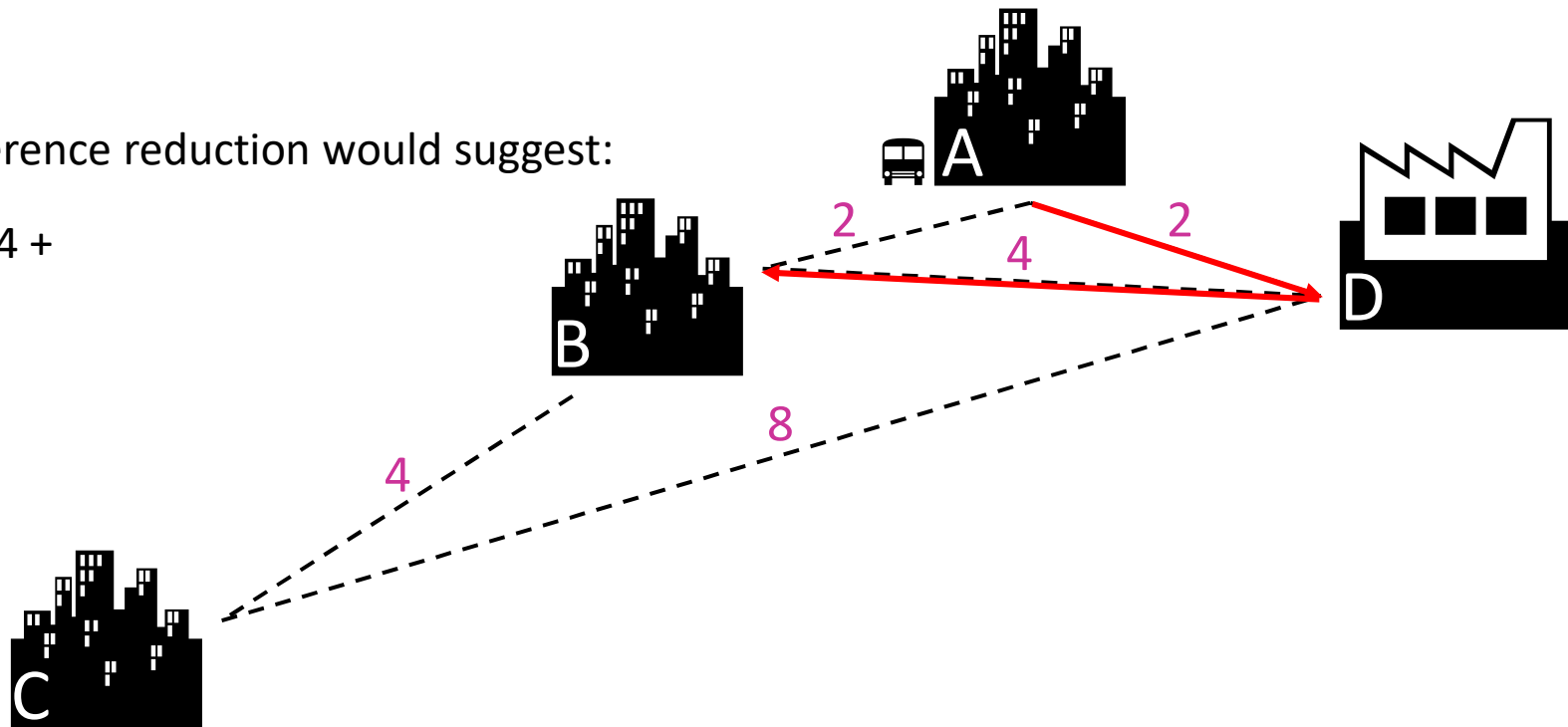


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

2 + 4 +

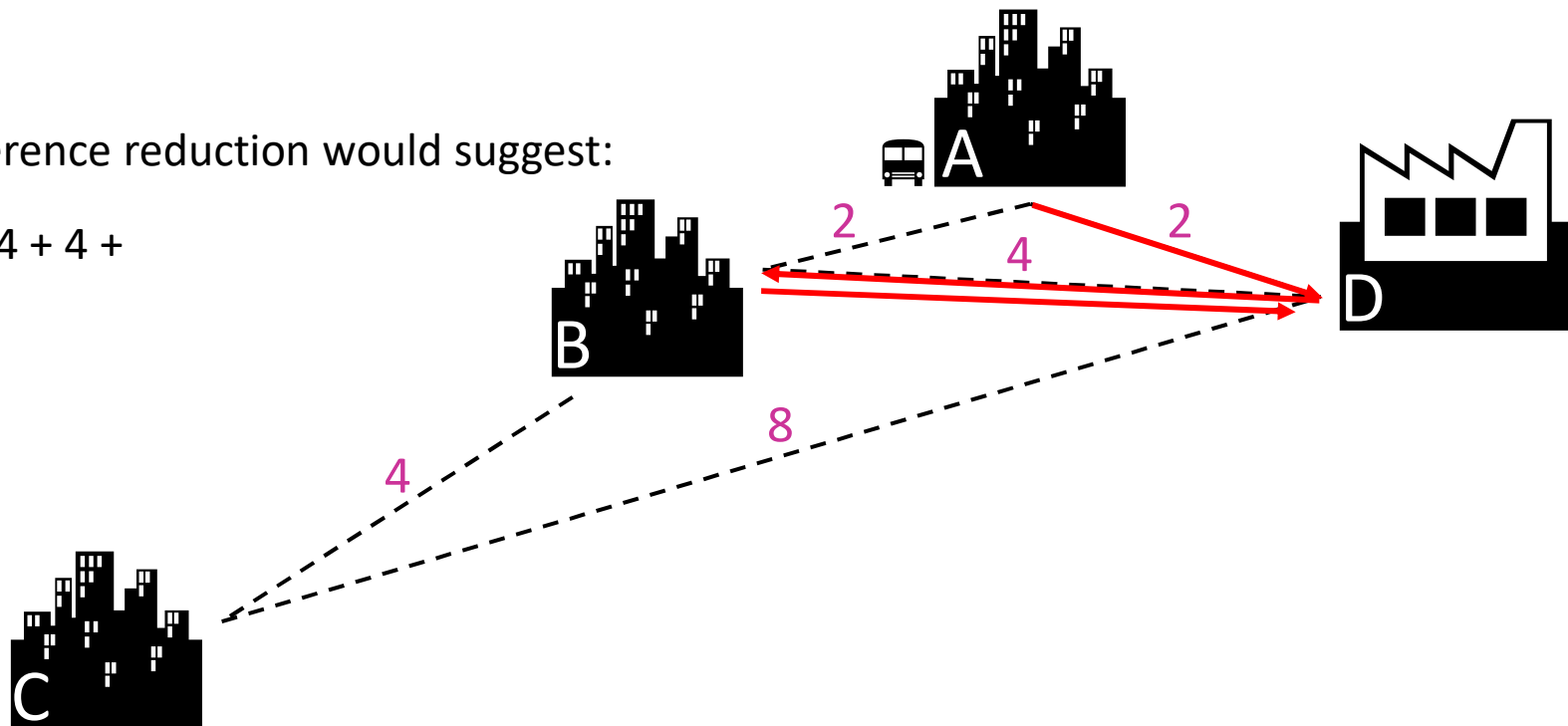


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

2 + 4 + 4 +

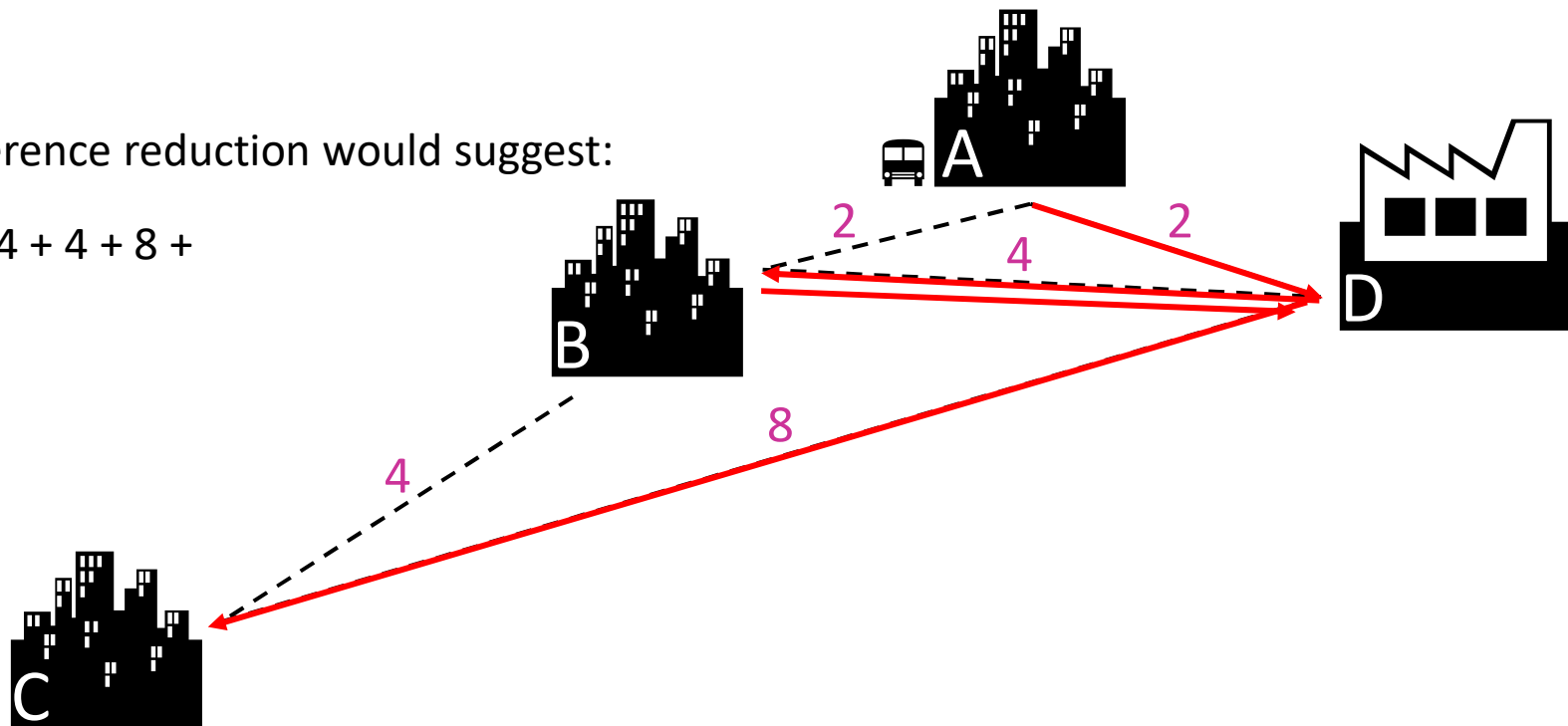


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$2 + 4 + 4 + 8 +$

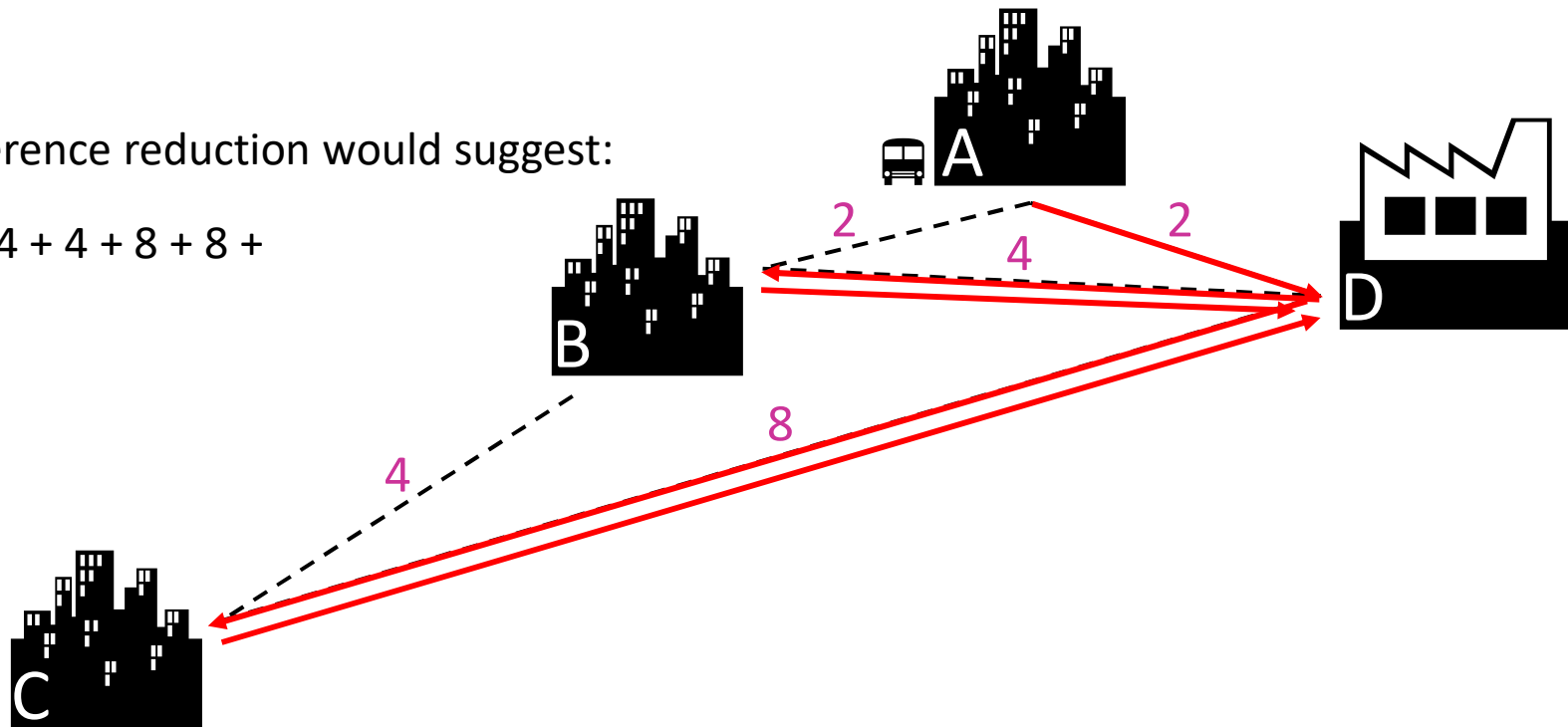


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$2 + 4 + 4 + 8 + 8 +$



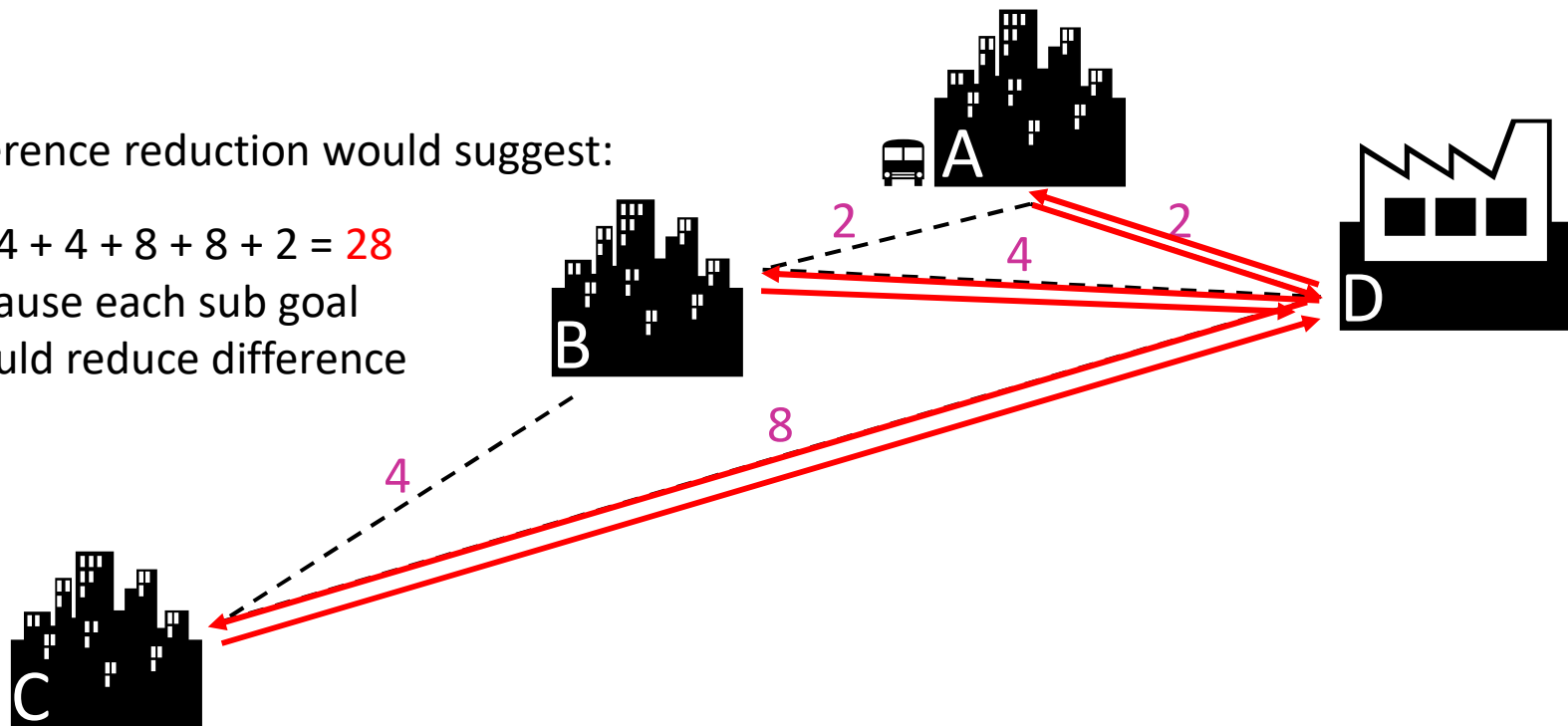
Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$$2 + 4 + 4 + 8 + 8 + 2 = 28$$

because each sub goal should reduce difference

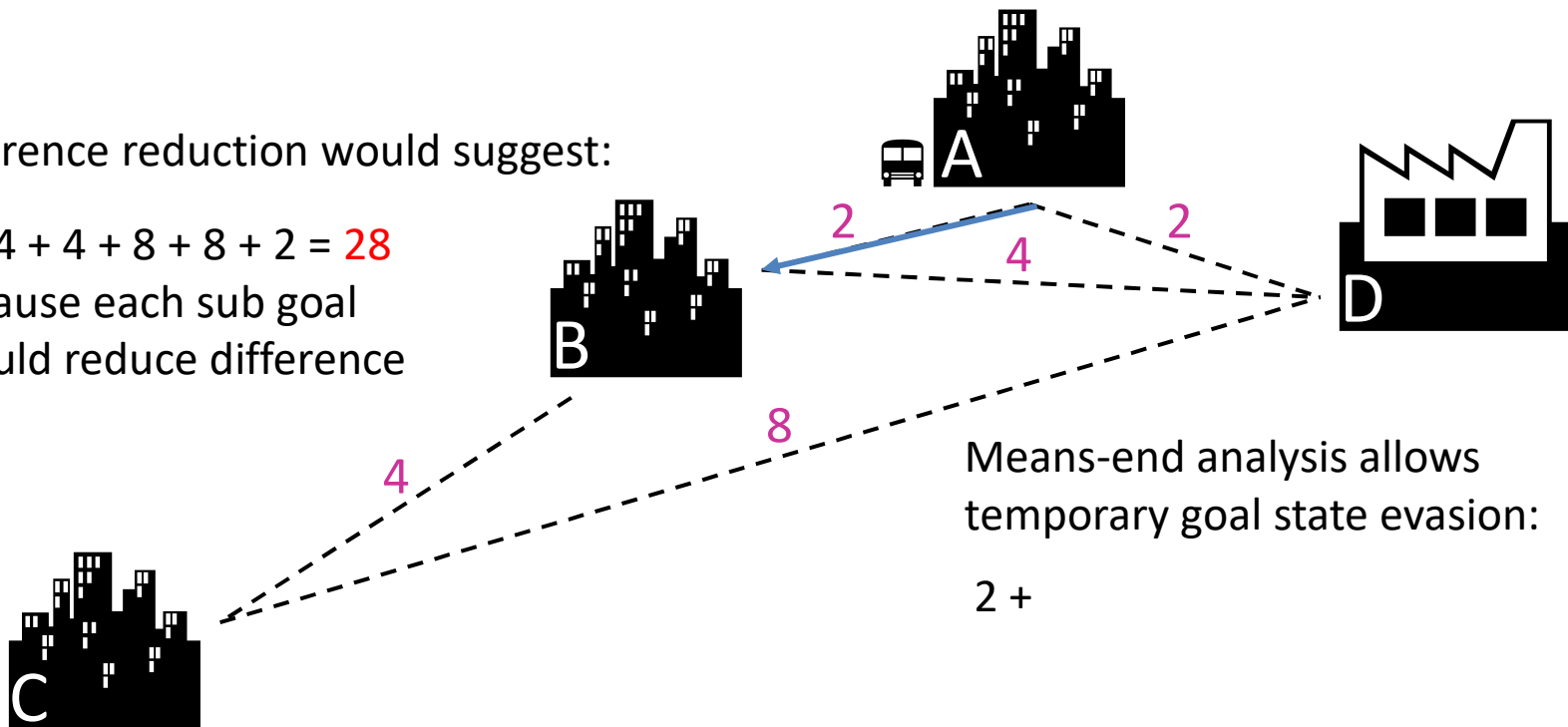


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because each sub goal
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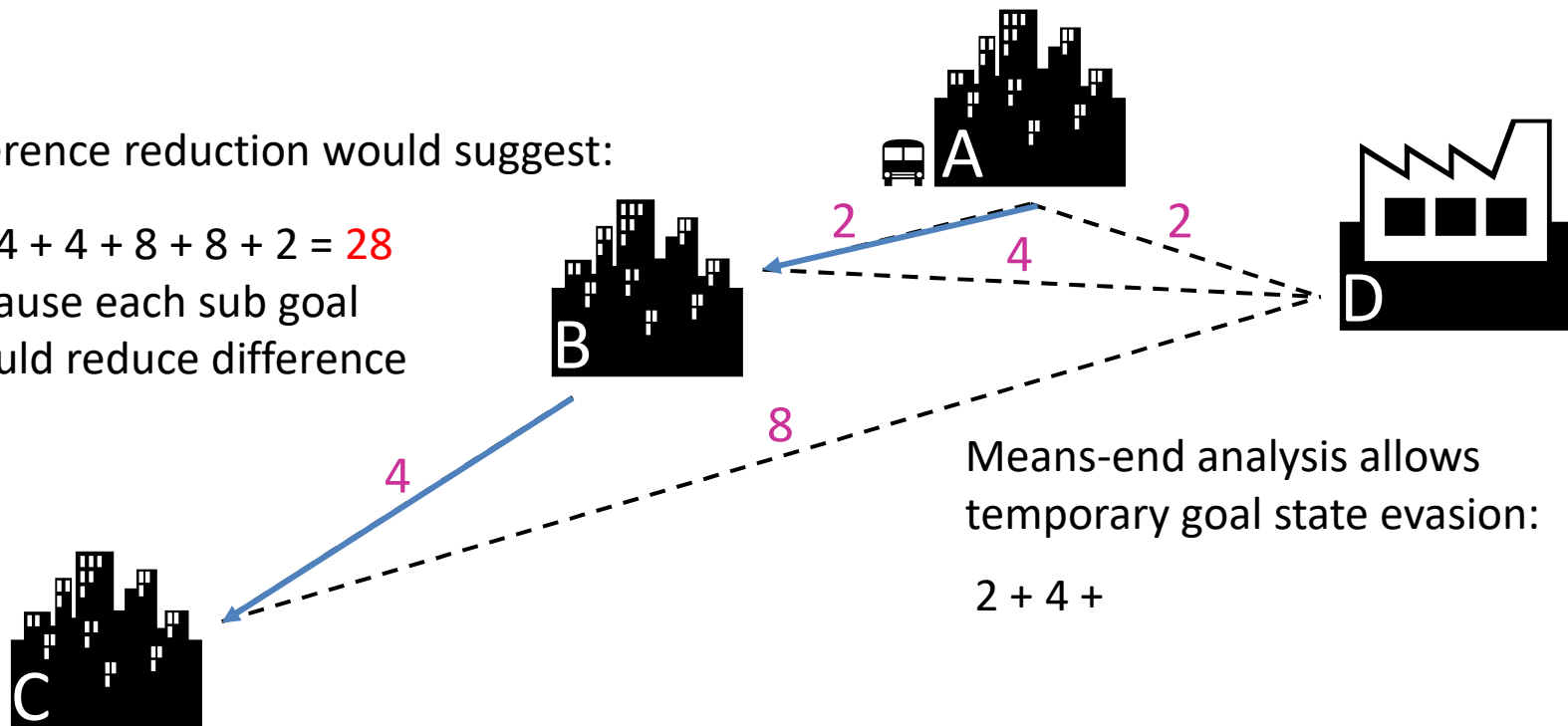


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$2 + 4 + 4 + 8 + 8 + 2 = 28$
because each sub goal
should reduce difference



Means-end analysis allows
temporary goal state evasion:

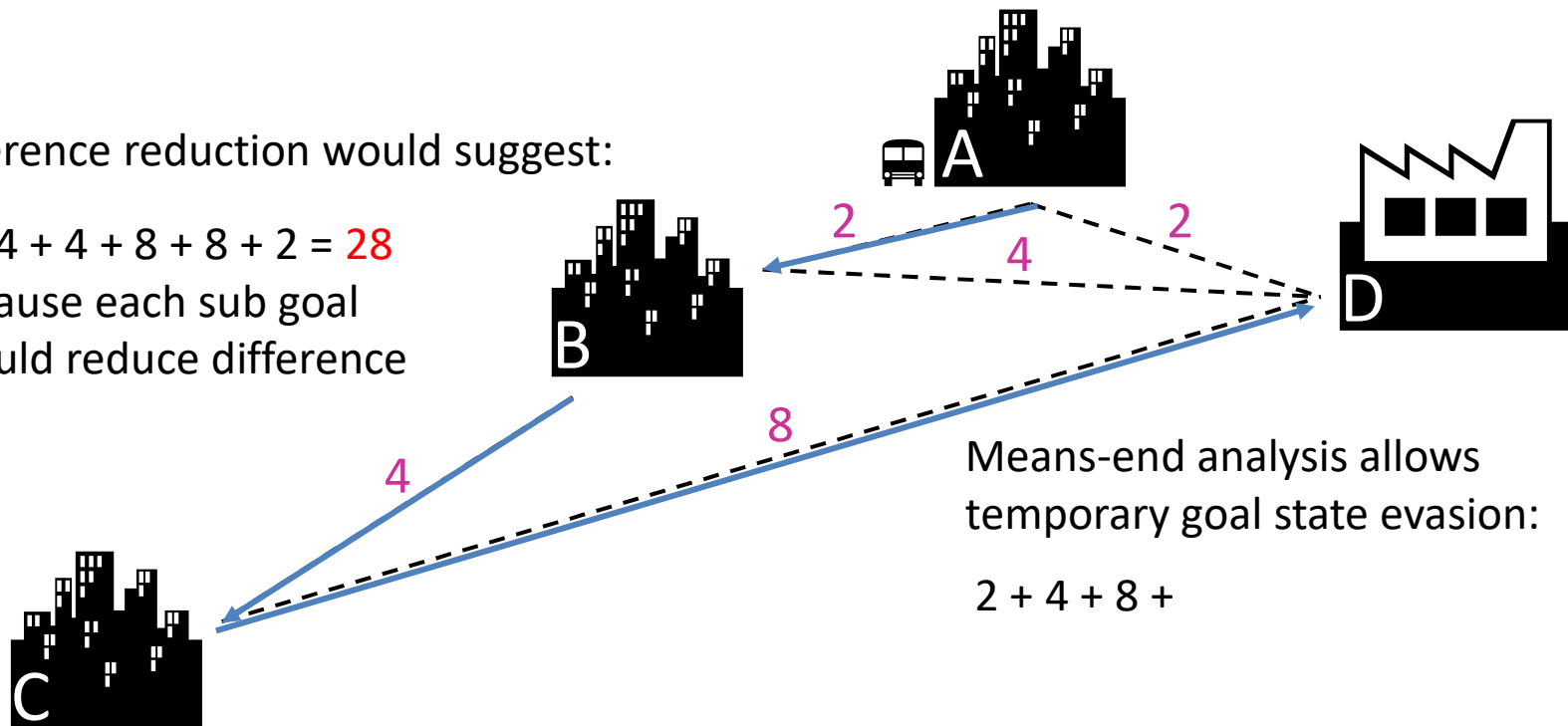
$2 + 4 +$

Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$2 + 4 + 4 + 8 + 8 + 2 = 28$
because each sub goal
should reduce difference

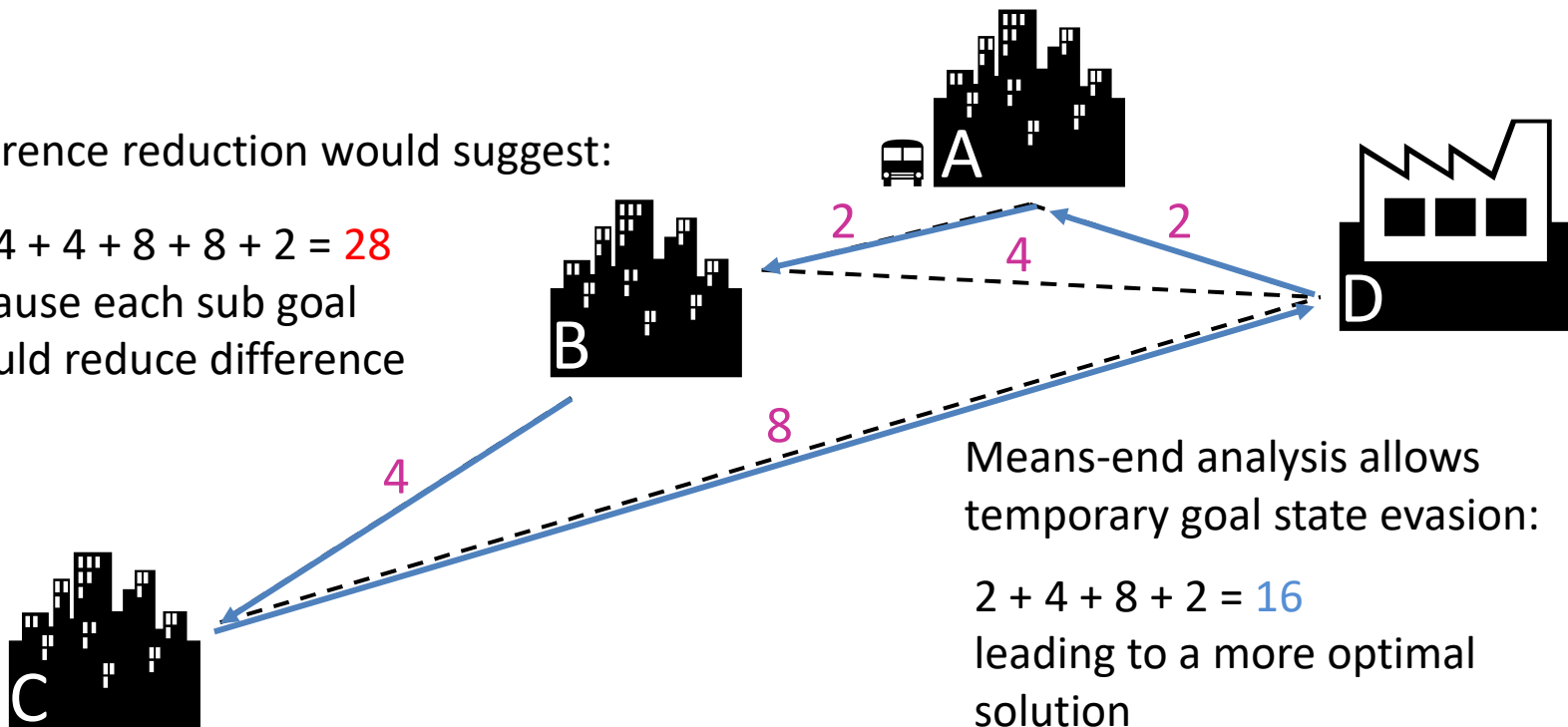


Means-end analysis: for each sub goal, a (different) procedure decreases the difference between current problem state and goal state even if this temporarily leads away from the goal state

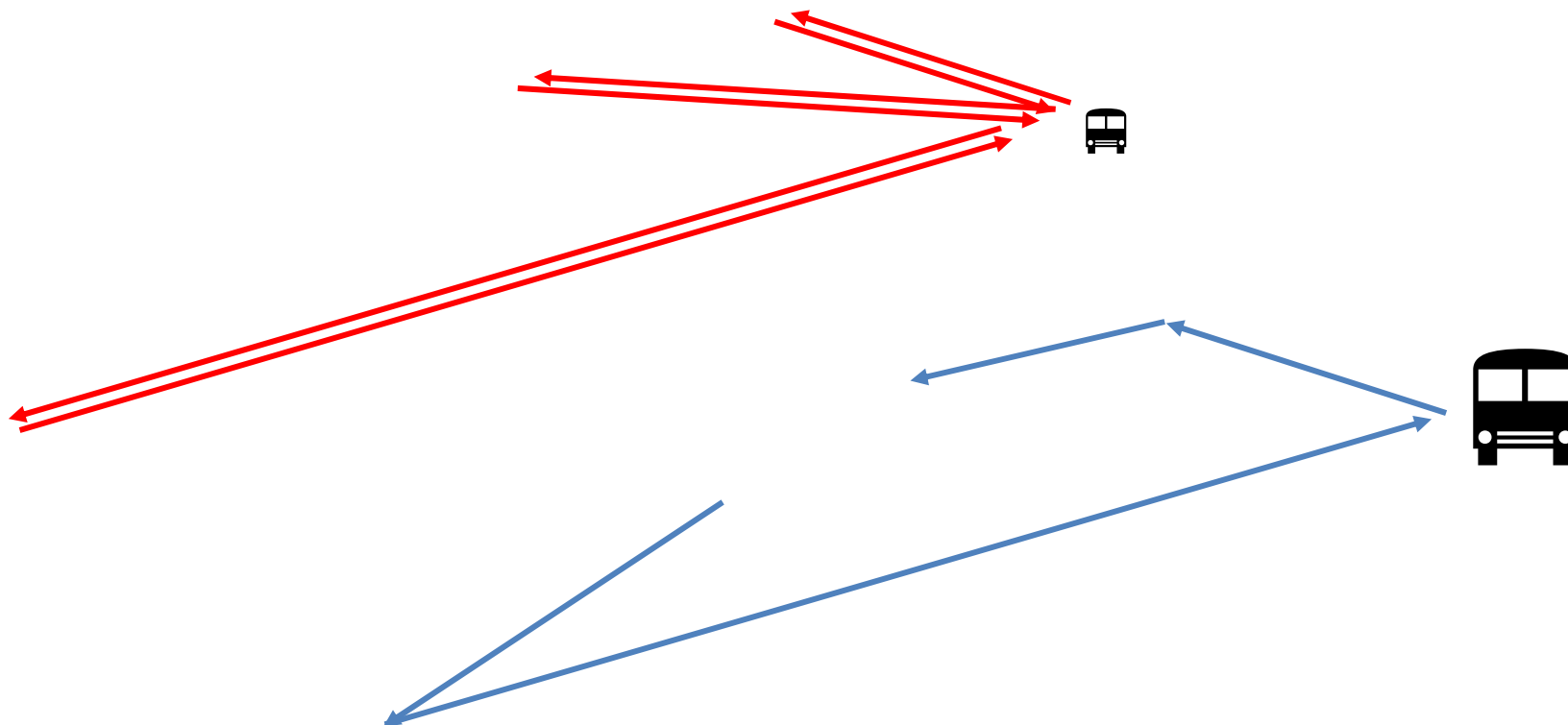
Logistic problem: supply factory from three different cities with one lorry, then return to base

Difference reduction would suggest:

$2 + 4 + 4 + 8 + 8 + 2 = 28$
because each sub goal
should reduce difference



Note: The choice of a problem-solving strategy takes effect on the technology you buy / develop / apply. In our case, difference reduction requires a smaller lorry but wear-and-tear is bigger (makes more miles); means-end analysis demands a larger lorry to transport all goods at once!!



Fetch robotics: Difference reduction or Means-end analysis?



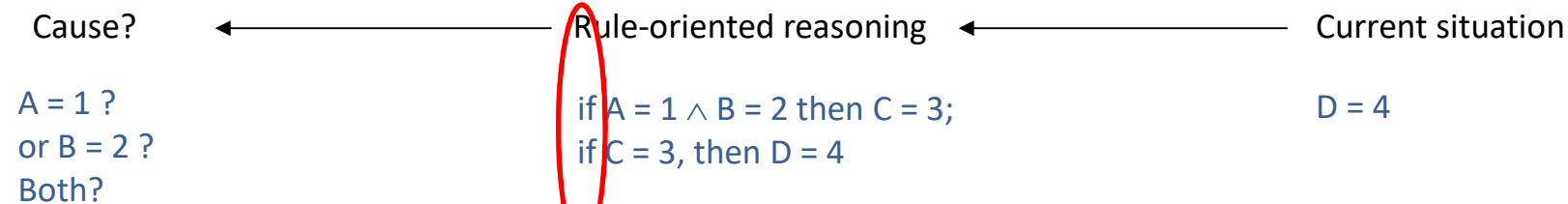
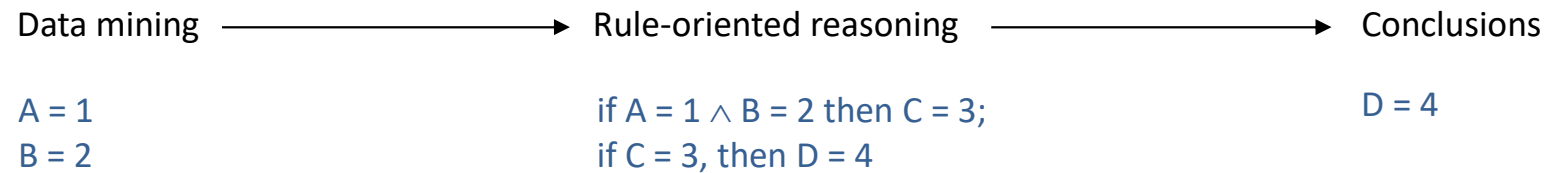
Difference reduction:
Means-end analysis:

more wear-and-tear
bigger robots / platforms

Forward and backward reasoning are recursive inference principles. Used to solve well-defined problems, which explicitly have formal limitations (chess, maths)

lonely = 1; socially isolated = 2; need for relationship = 3; social robot = 4; social media = 5; dating site = 6

Forward reasoning



Backward reasoning

Depends on formal relationship between variables. If there are more explanations for the same phenomenon, see next slide...

Forward and backward reasoning are recursive inference principles. Used to solve well-defined problems, which explicitly have formal limitations (chess, maths)



Your PC ran into a problem and needs to restart. We're just collecting some error info, and then we'll restart for you.

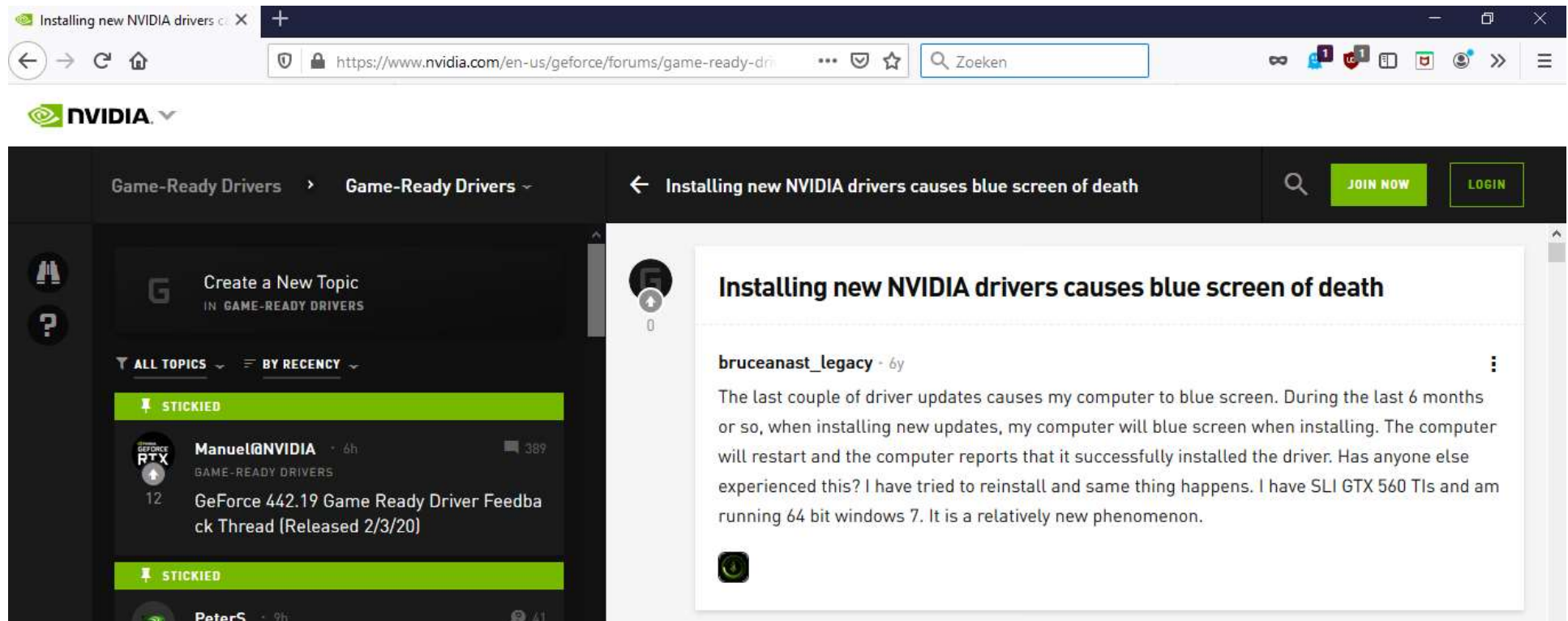
42% complete



For more information about this issue and possible fixes, visit
<https://www.windows.com/stopcode>

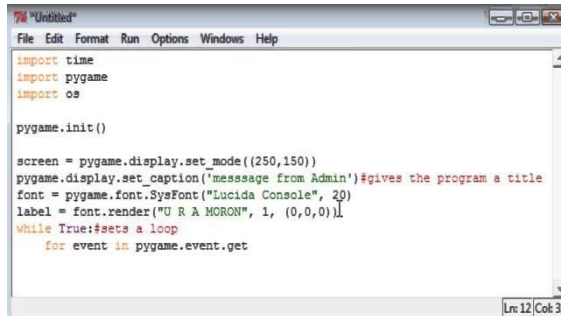
If you call a support person, give them this info:
Stop code: PAGE_FAULT_IN_NONPAGED_AREA
What failed: nvlddmkm.sys

If it is a known problem...
(I just installed that new NVIDIA driver)



... you are allowed to reason forward and in this case, backward
(diagnostics)

If the problem is unknown...
(more explanations for the same phenomenon)

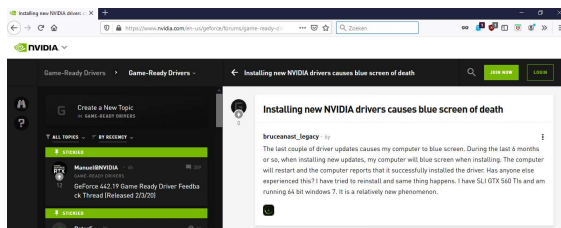


```
import time
import pygame
import os

pygame.init()

screen = pygame.display.set_mode((250,150))
pygame.display.set_caption('message from Admin')#gives the program a title
font = pygame.font.SysFont("Lucida Console", 20)
label = font.render("U R A MORON", 1, (0,0,0))
while True:#sets a loop
    for event in pygame.event.get
```

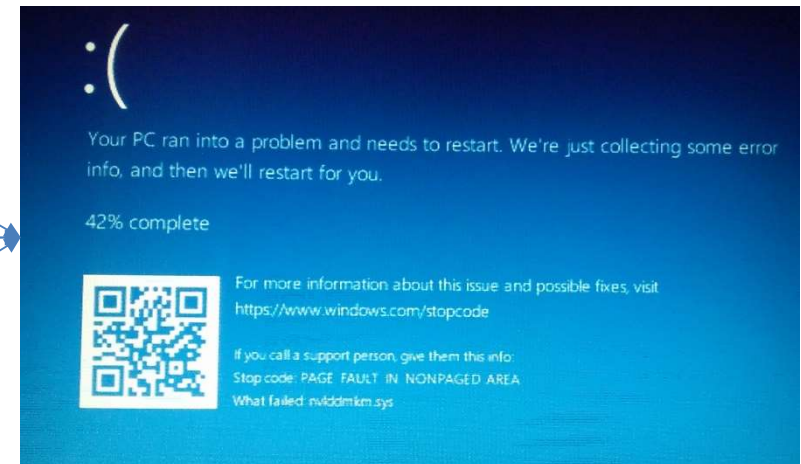
"I was coding a virus"



"Installed NVIDIA driver"



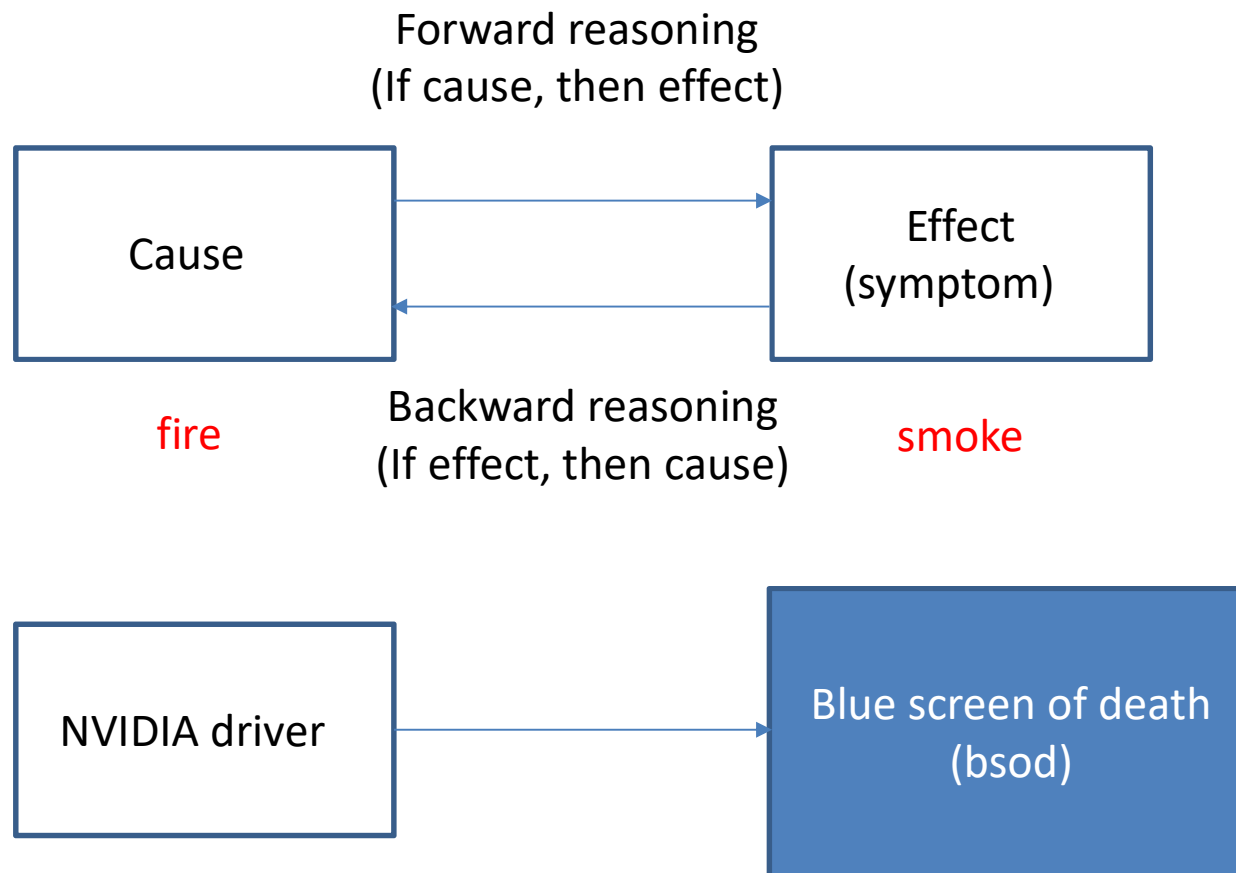
"I gamed so hard,
I overheated my CPU"



... you are NEITHER allowed to reason forward NOR backward
without empirical validation, excluding all causes but 1

Logic reasoning

Only allowed for known problems
(must have 1 to 1 relationship)
Else, guesstimate and test empirically



Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”

Features: clock signal = heartbeat

Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

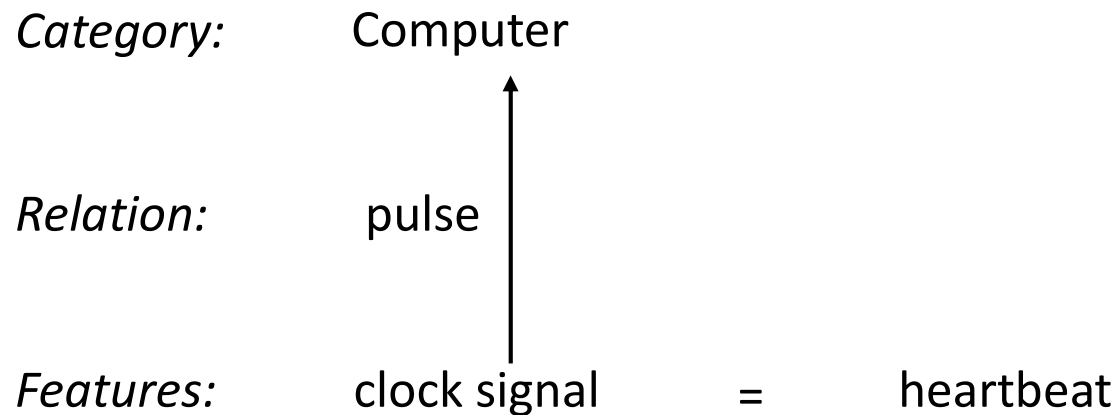
UNIX manual: “The clock signal is the heartbeat of your computer”

Category: Computer

Features: clock signal = heartbeat

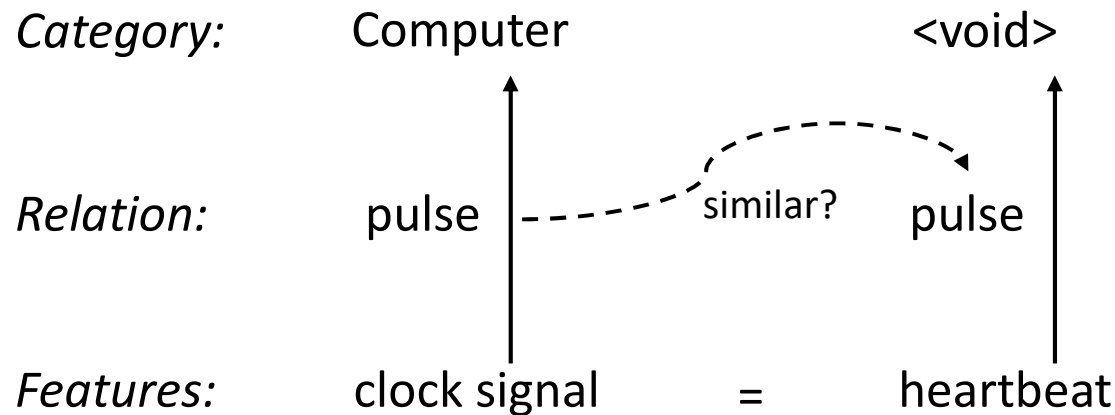
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”



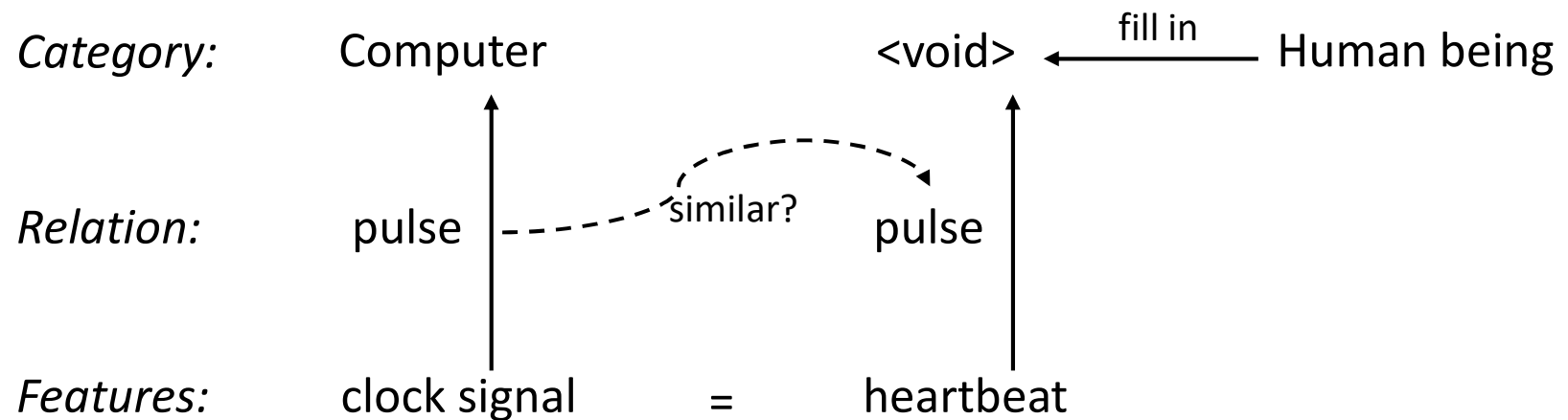
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”



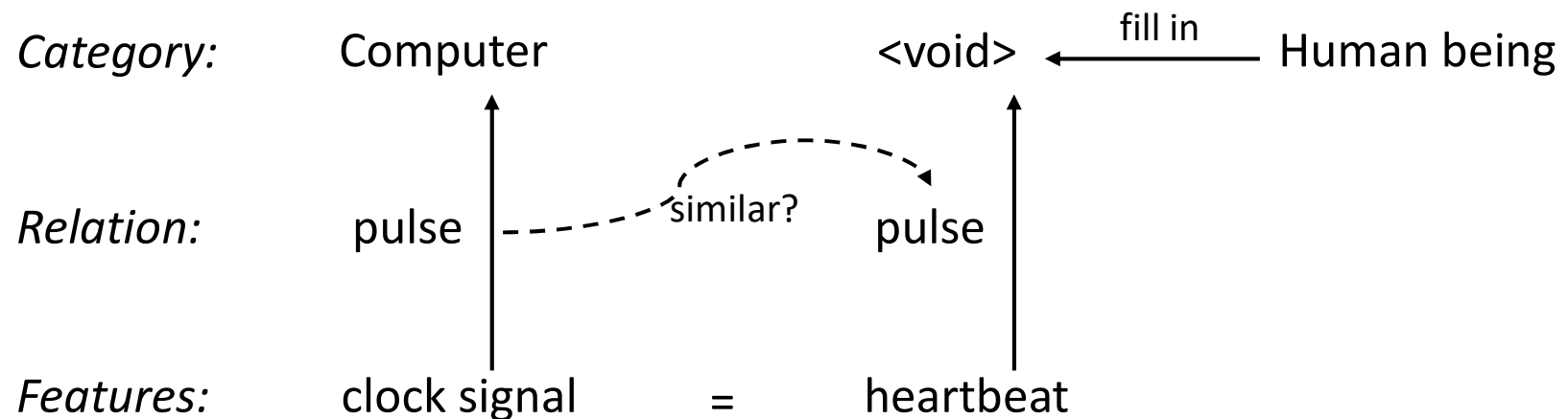
Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

UNIX manual: “The clock signal is the heartbeat of your computer”



Analogical reasoning looks for relations between features and categories that are similar to relations between other features and categories

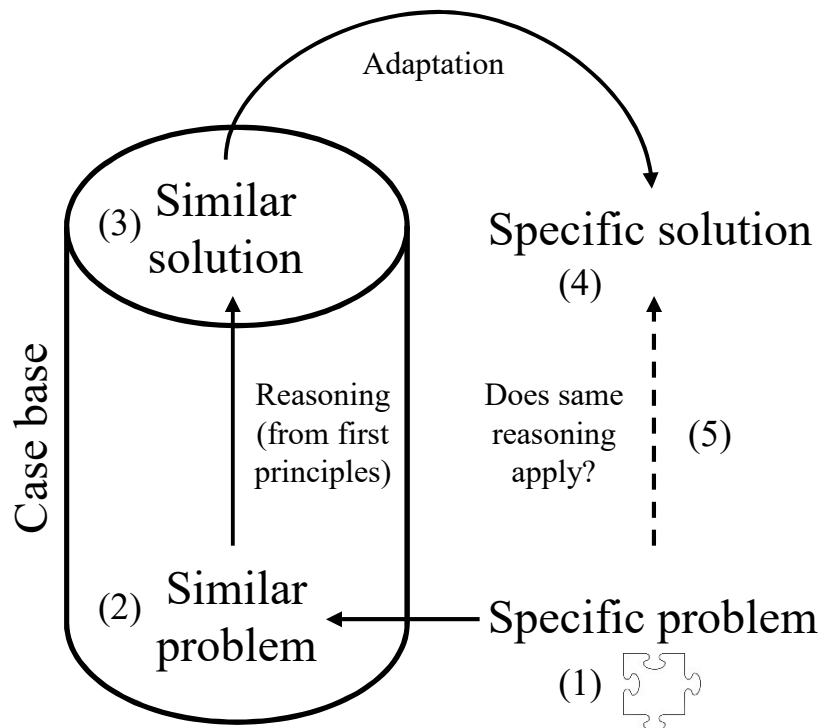
UNIX manual: “The clock signal is the heartbeat of your computer”



Clock signal : computer :: heartbeat : human being

Analogical reasoning is halfway intelligent and creative problem solving: It has a reasoning and an associative side

Case-based reasoning is a type of analogical reasoning



- 1) Axe imbalance and inefficiency
- 2) Who in nature works like an axe? (association)
- 3) How does he do it?
- 4) Adapt woodpecker neck form to fit my camp axe
- 5) It is probably for the same reason that it works so well (energy absorption)

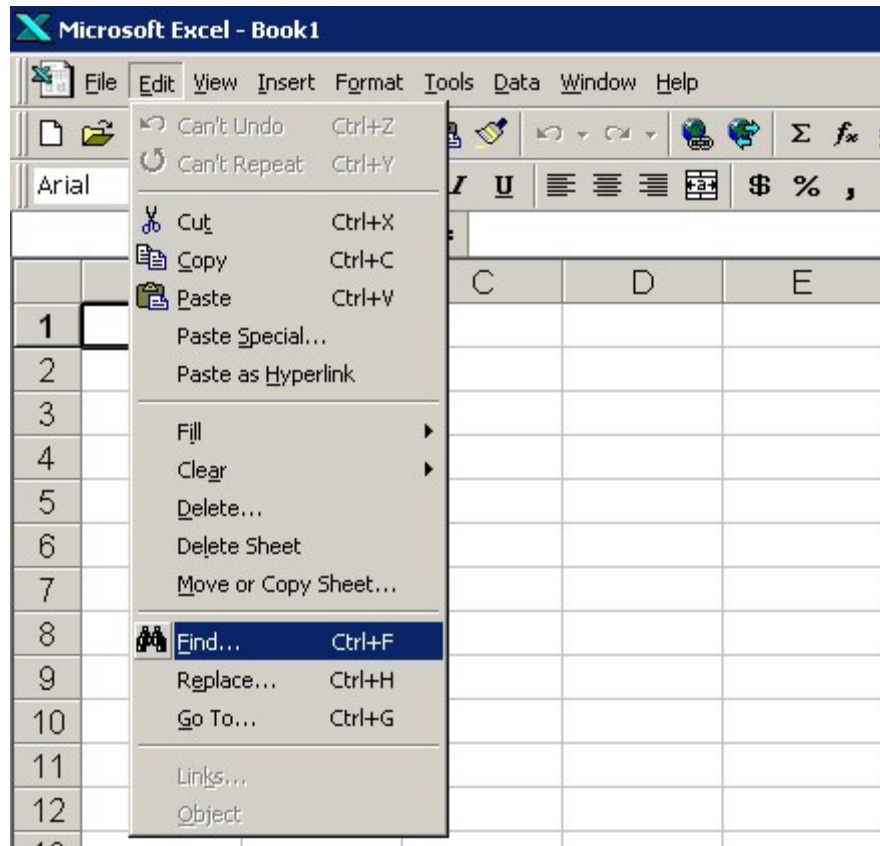
The creative combination is in the abstraction of finding similar/same principles in otherwise unrelated domains



Woodpecker-inspired camp axe for more balanced swing and more efficient blow

Old solution : old problem :: adapted solution : new problem

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



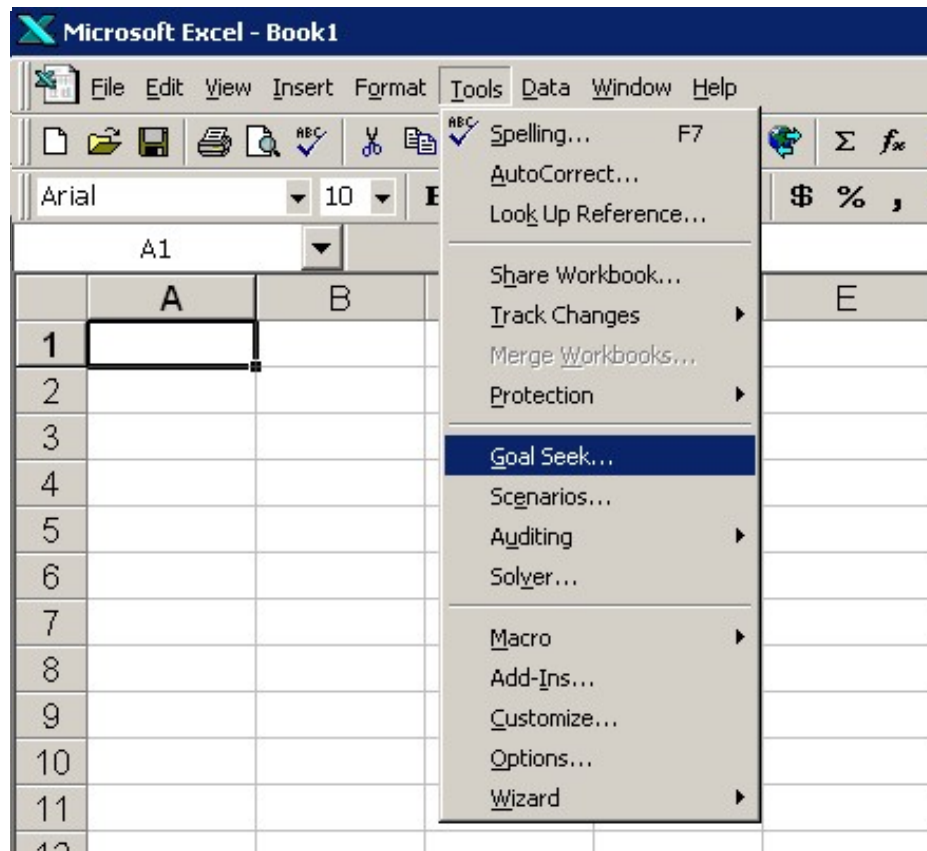
Novices follow first association.

Their problem:

Find (Ctrl F) is not related in meaning to Edit

Violates the demand in intelligent problem-solving of systematic analysis and planned action. In creative problem-solving, serendipitous findings may occur

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



Novices follow first association.

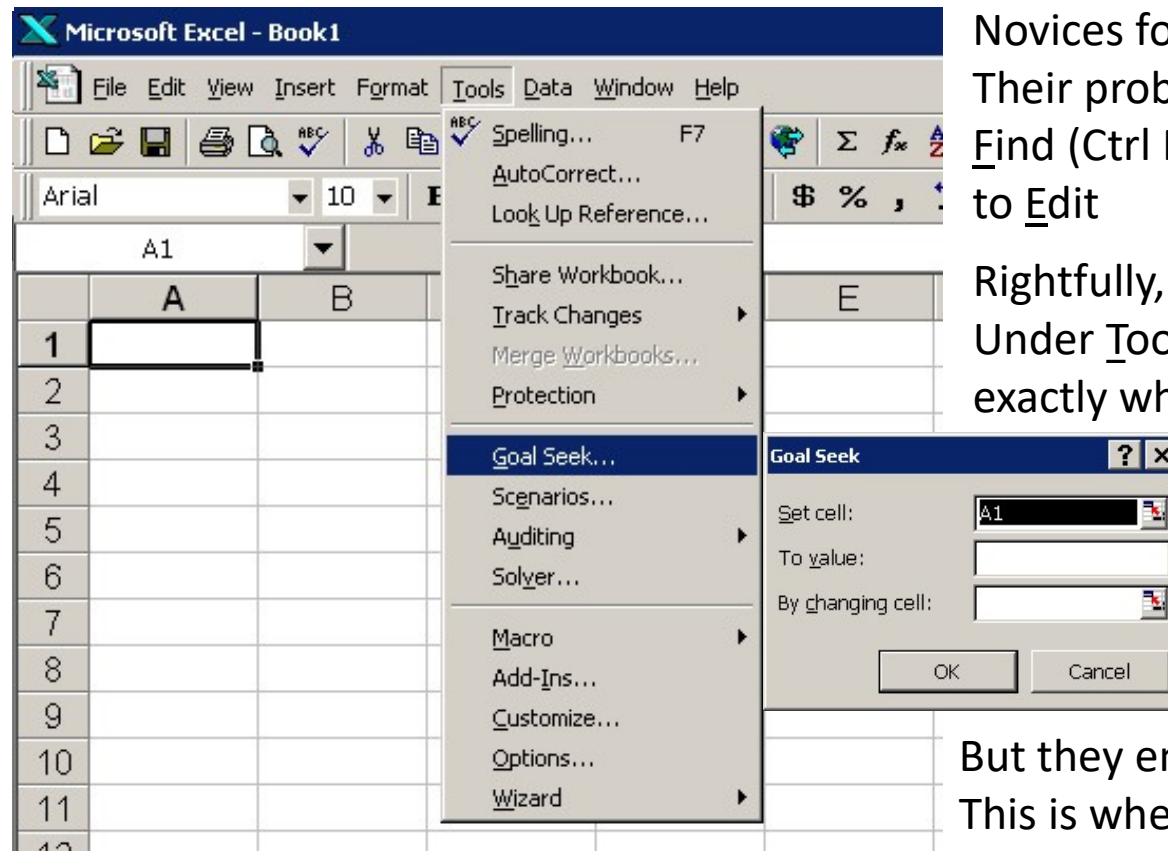
Their problem:

Find (Ctrl F) is not related in meaning to Edit

Rightfully, they see searching as a tool. Under Tools, Goal Seek... describes exactly what they want to do

Violates the demand in intelligent problem-solving of systematic analysis and planned action. In creative problem-solving, serendipitous findings may occur

Random trial-and-error is a low level problem solving strategy but commonly found with non-expert users



Novices follow first association.

Their problem:

Find (Ctrl F) is not related in meaning to Edit

Rightfully, they see searching as a tool. Under Tools, Goal Seek... describes exactly what they want to do

But they end up setting cells to values
This is where the wild clicking commences...

Violates the demand in intelligent problem-solving of systematic analysis and planned action. In creative problem-solving, serendipitous findings may occur

Problem solving strategies

Intelligence

Rule following
Difference reduction

Means-end analysis
Forward and backward reasoning

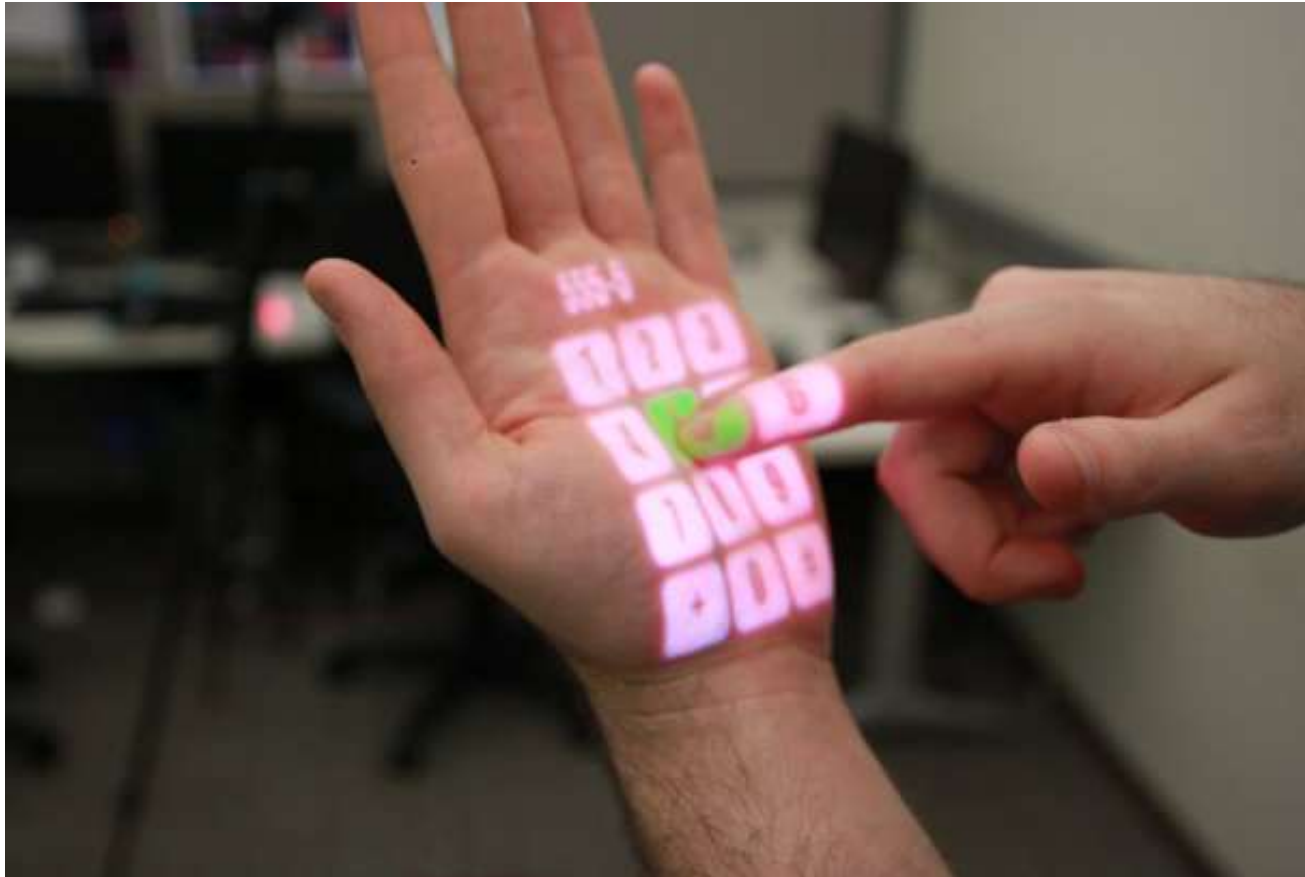
Creativity

Analogy
Random trial and error

Creative problem solving

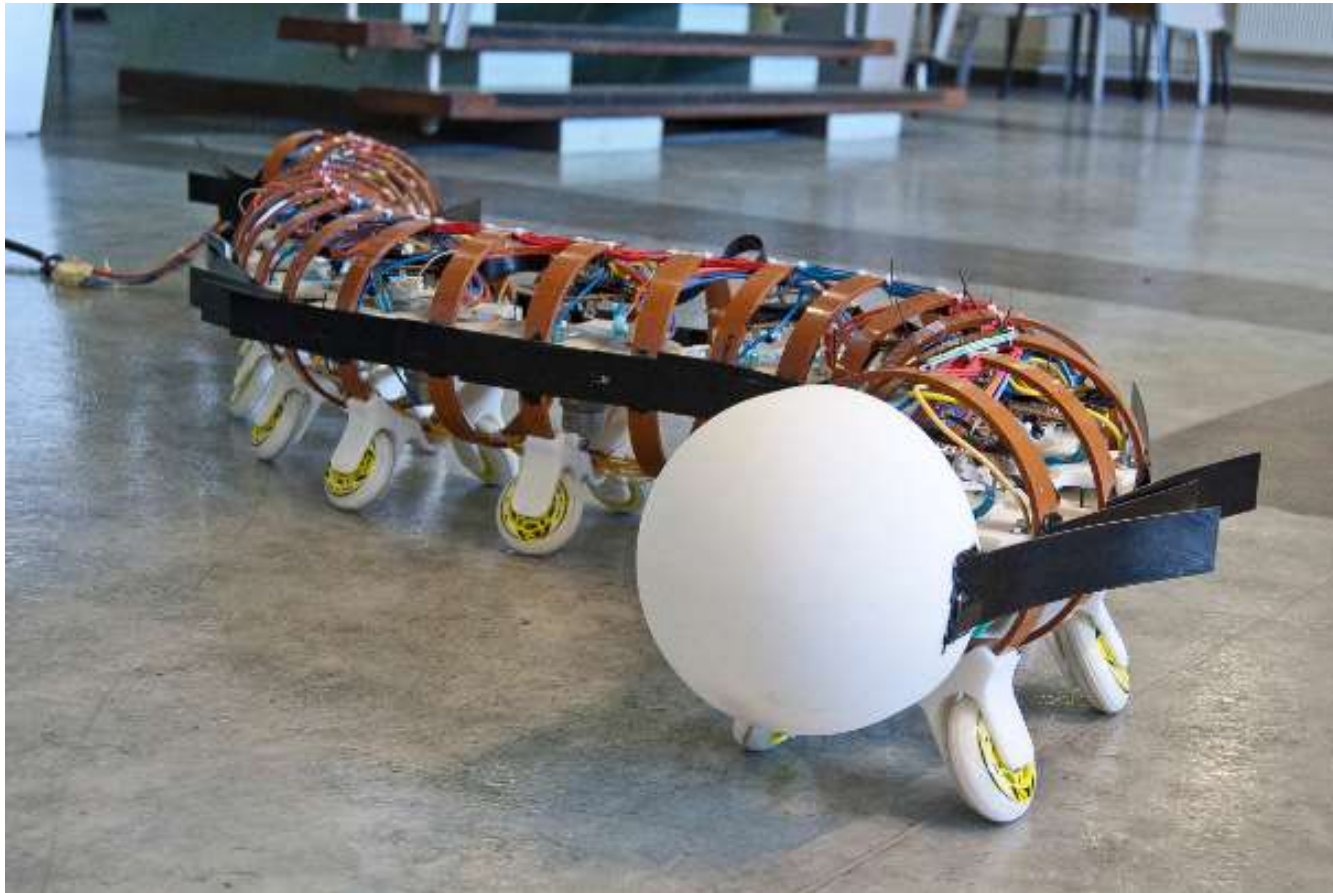
May lead to innovation

Creative problem solving: give a different function to the (standard) means that are available to solving the problem



OmniTouch augmented reality. Turns any surface - from a table to a hand - into a touchscreen. Wearable device that projects a graphical user interface, such as a computer desktop or a virtual keypad, onto any surface. Users are able to interact with these virtual interfaces using their fingers, which are tracked by sensors built into OmniTouch, in the same way they would use a touchscreen (Nick Heath, 2011).

Creative problem solving: The fusion of two worlds
(e.g., biology (worm) and engineering (rescue robot))



University of Leeds (Jordan Boyle). Robot movement based on locomotion of real-life nematode worms. Possible applications: Searching for survivors in collapsed buildings or navigating gaps in damaged structures to deliver aid to individuals trapped in disaster zones (Nick Heath, 2011).

Creative problem solving: Going beyond the obvious
(e.g., using the backside of a PC)



Crowd PC (Philipp Schaake). Runner-up prize in the Lifebook category of the Fujitsu Design Award 2011. Can be configured to resemble a notebook or slate, and can be tweaked so that the screen and keyboard are separate. The device's modifiable nature also allows it to be operated via the back cover when closed (Nick Heath, 2011).

Next Step: Creativity and Deep Learning (DL)

DL is a type of computer program (AI) from the family of neural networks

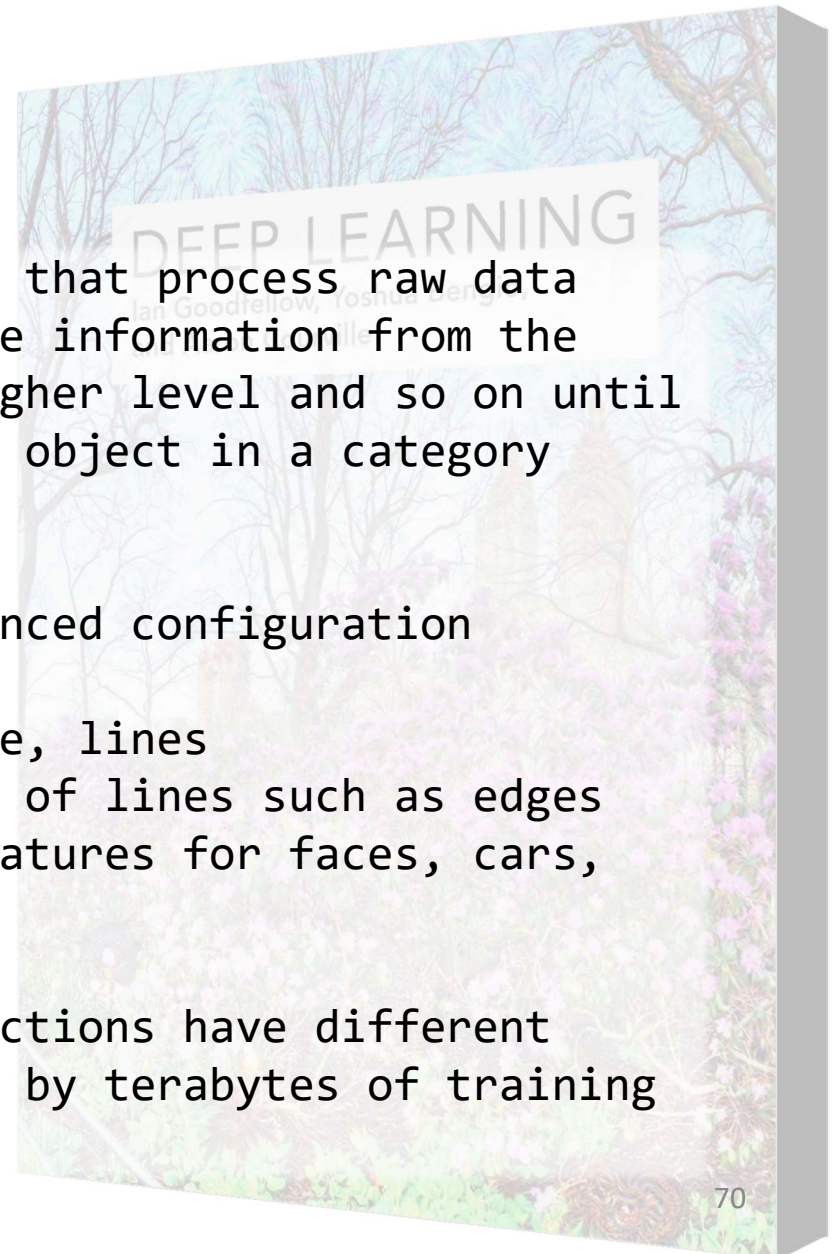
It has 10 to 20 layers of 'cells' that process raw data (e.g., one pixel). Cells integrate information from the bottom layers up into the next higher level and so on until they reach criterion and place an object in a category based on its features

DL is signal detection in an advanced configuration

Bottom layers detect, for instance, lines

Second layers search combinations of lines such as edges
Edges together make up typical features for faces, cars, elephants, chairs, etc.

All cells are connected but connections have different weights. Tweaking weights is done by terabytes of training data



Next Step: Creativity and Deep Learning (DL)

Which is the odd one out?



Weights indicate the relative importance of each cell

During training, the system gets thousands of pictures with the correct output it should produce

All weights are wrong but from the difference with the desired output the system adapts the weights to come closer (not this edge but that)

Through many many iterations, the weights are finetuned such that they produce 'fish' even in cases that were not in the training set. At that point, a robot with DL installed learns that you are a human and not a gorilla

(Courtesy GooglePhotos)

Next Step: Creativity and Deep Learning (DL)

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DIGITS

Google Mistakenly Tags Black People as 'Gorillas,' Showing Limits of Algorithms

By **ALISTAIR BARR**
Jul 1, 2015 3:40 pm ET

Google is a leader in artificial intelligence and machine learning. But the company's computers still have a lot to learn, judging by a major blunder by its Photos app this week.

The app tagged two black people as "Gorillas," according to Jacky Alciné, a Web developer who spotted the error and tweeted a photo of it.

Recommended Videos

1. Film Clip: 'Lost in Paris'
2. Film Clip: 'Cars 3'

Through many many iterations, the weights are finetuned such that they produce 'fish' even in cases that were not in the training set. At that point, a robot with DL installed learns that you are a human and not a gorilla

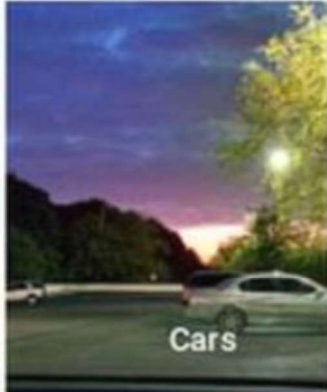
(Courtesy GooglePhotos)

Next Step: Creativity and Deep Learning (DL)

 **diri noir avec banan** @jackyalcine · Jun 28
Google Photos, y'all [redacted] up. My friend's not a gorilla.

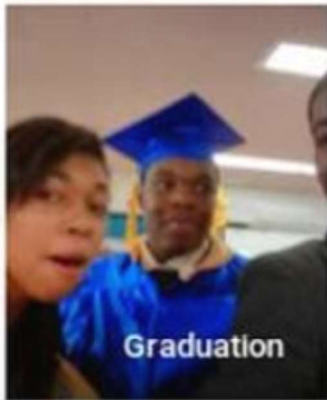
 Skyscrapers

 Airplanes

 Cars

 Bikes

 Gorillas

 Graduation

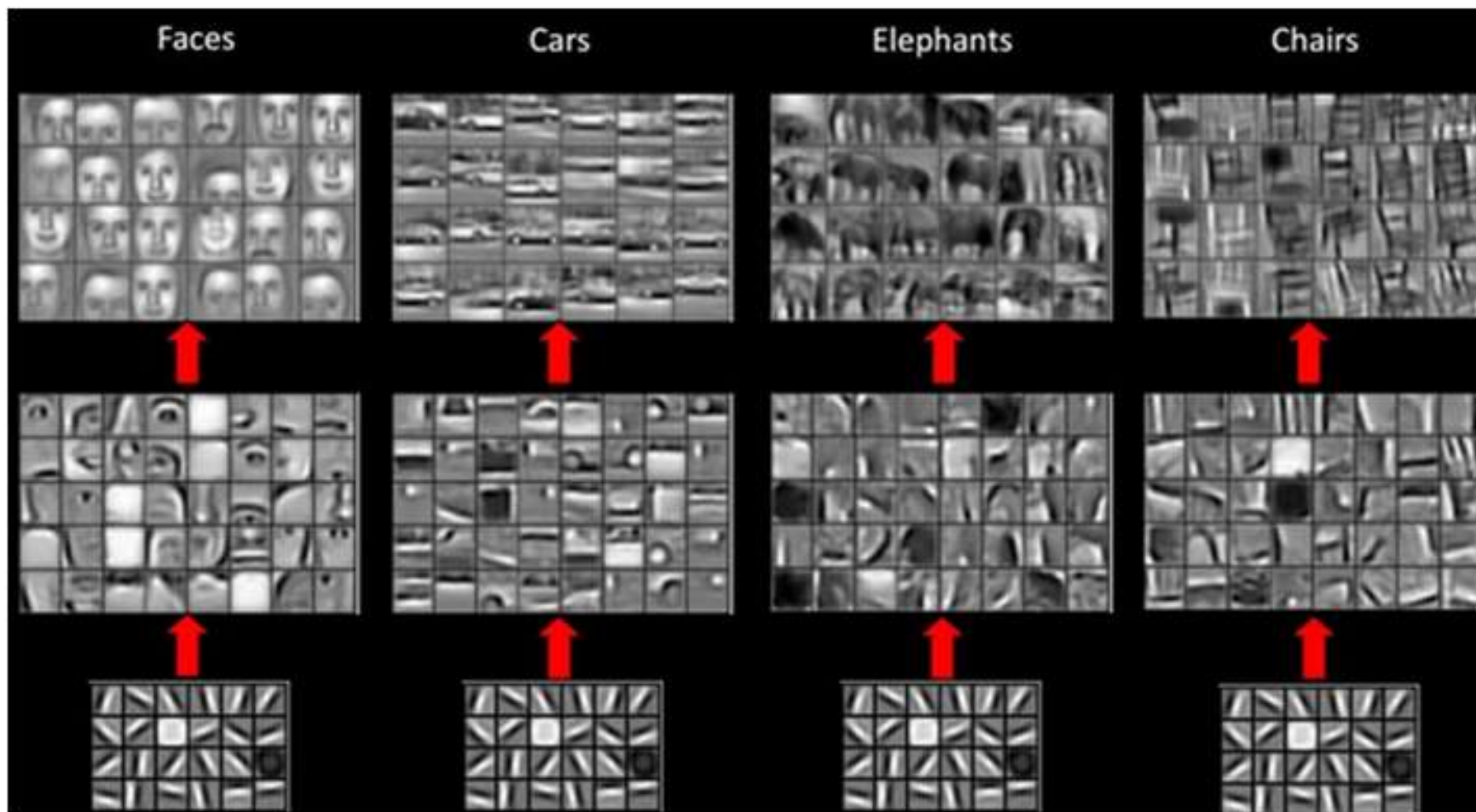
RETWEETS 1 031 FAVORITES 513



Why error is only error in a rigid system and not in creativity

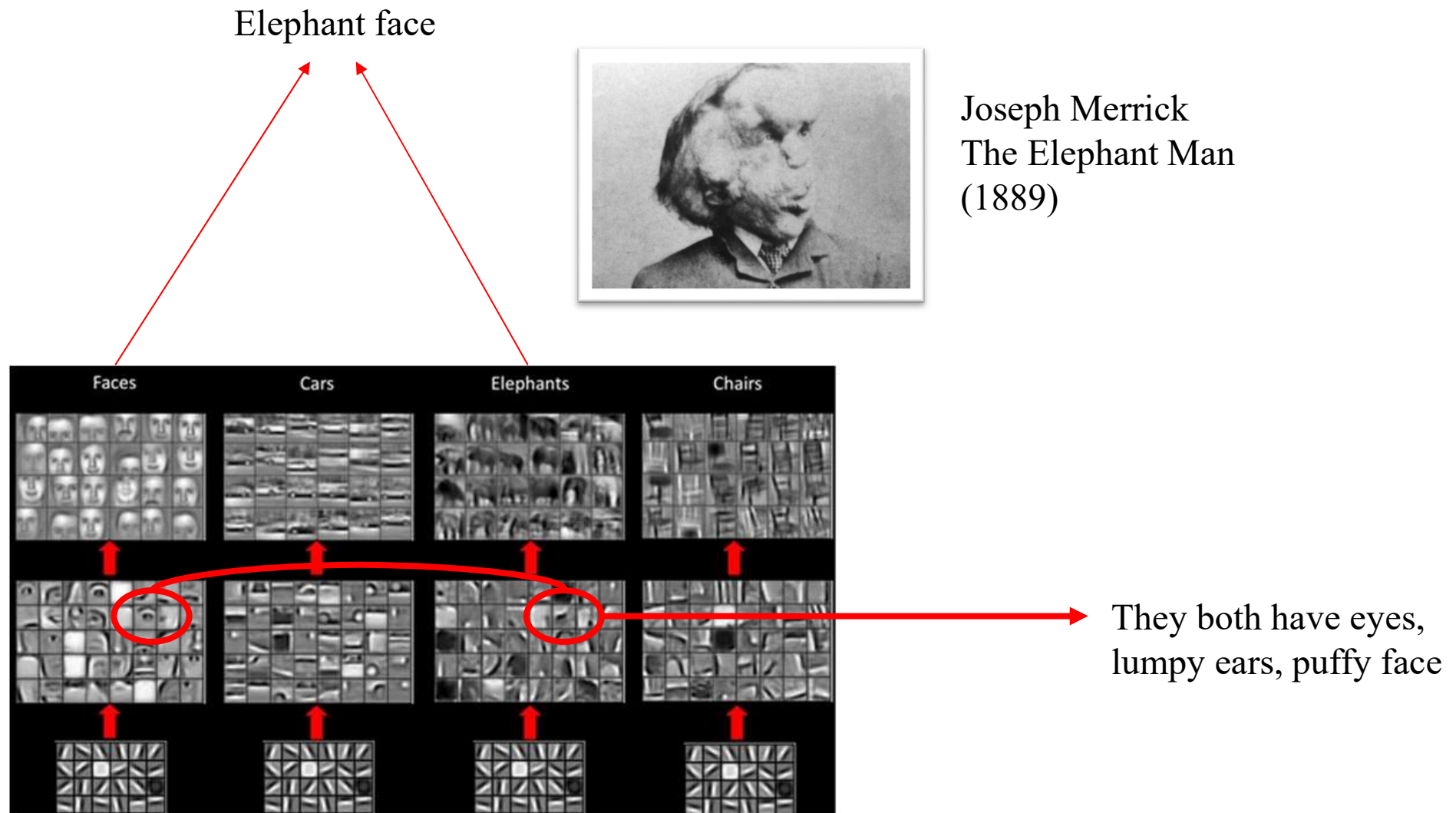
The case of mistakes in Deep Learning

Categorization from the bottom features up



Why error is only error in a rigid system and not in creativity

Features from the 'wrong' category may match (sometimes on an abstract level)



Why error is only error in a rigid system and not in creativity

Features from the 'wrong' category may match (sometimes on an abstract level)

Car face

Faces Cars Elephants

quora.com/Why-does-a-car-front-look-like-a-face

Quora Search for questions, people, and topics

Why does a car front look like a face?

Nick Schön
Car lover all my life, classic car owner and tinkerer · Author has 6.4K answers and 40.6M answer views · 2y

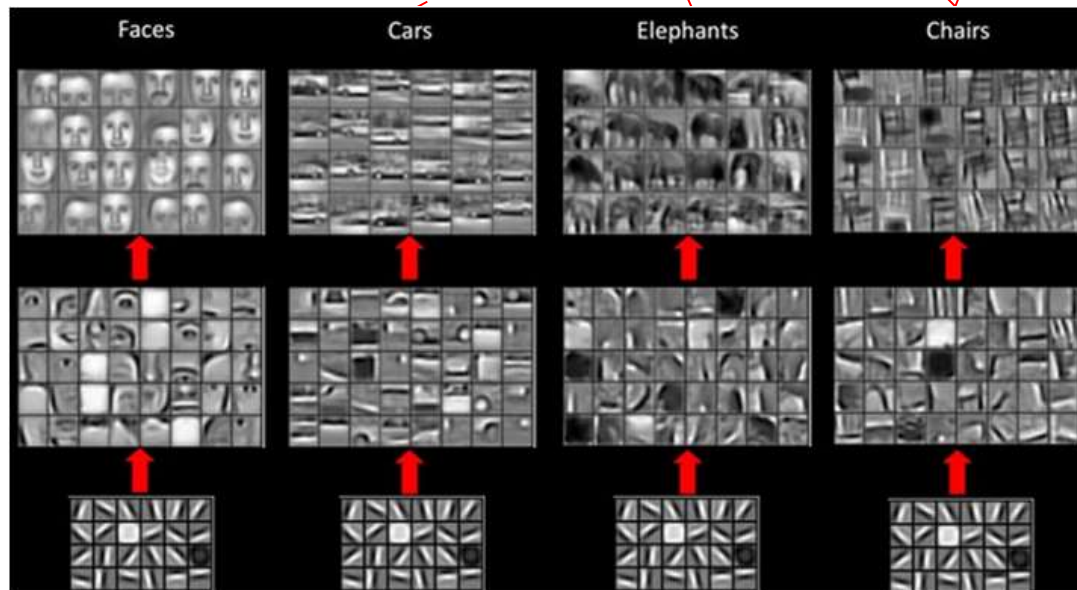
Because automotive designers do that on purpose

Why error is only error in a rigid system and not in creativity

Hybrids exist, maybe more than NN classifiers wish for

Elephant chair!

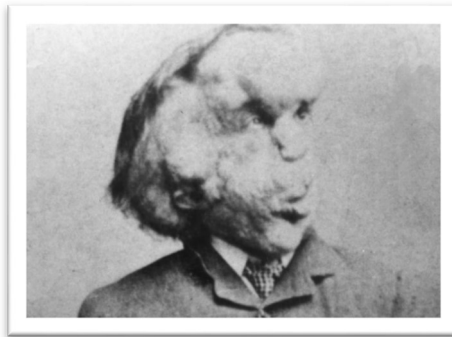
Car chair (kinda obvious)



Maximo Riera's Elephant Chair

Subsidiary conclusion: DL may not be all that wrong
when creativity is a result of information processing in general

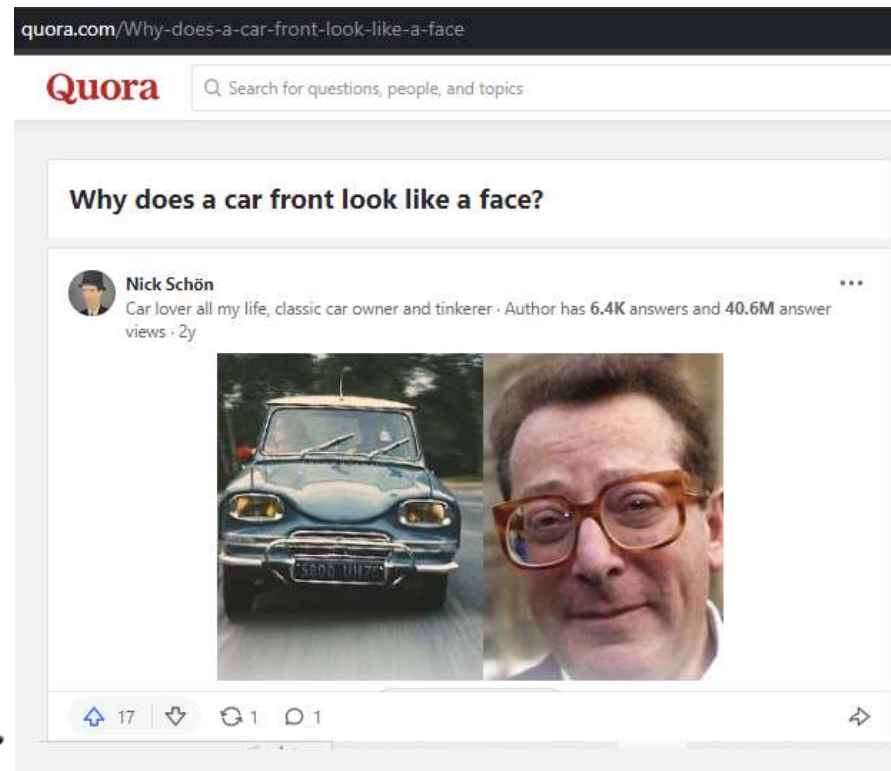
It's just that we want our NNs to be precise where they can't be

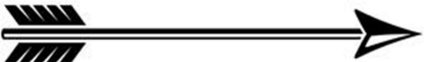


The Elephant Man



Elephant Chair



Humans do it all the time: 

Why error is only error in a rigid system and not in creativity

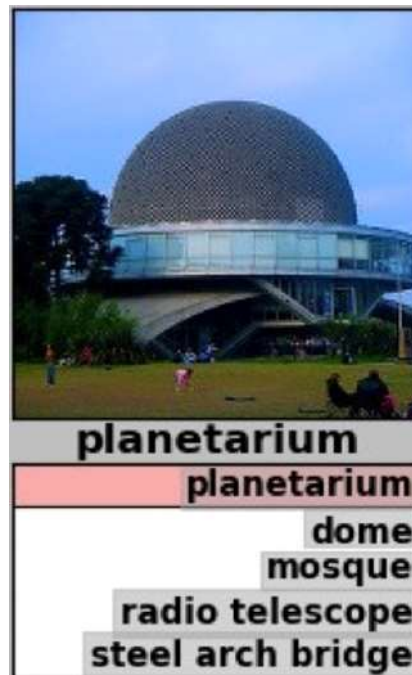
Features from the ‘wrong’ category may match (sometimes on an abstract level)

Mistake?



Machine-made

Mistake?



Machine-made

Not a mistake?

Why isn't this a pile of vegetables?



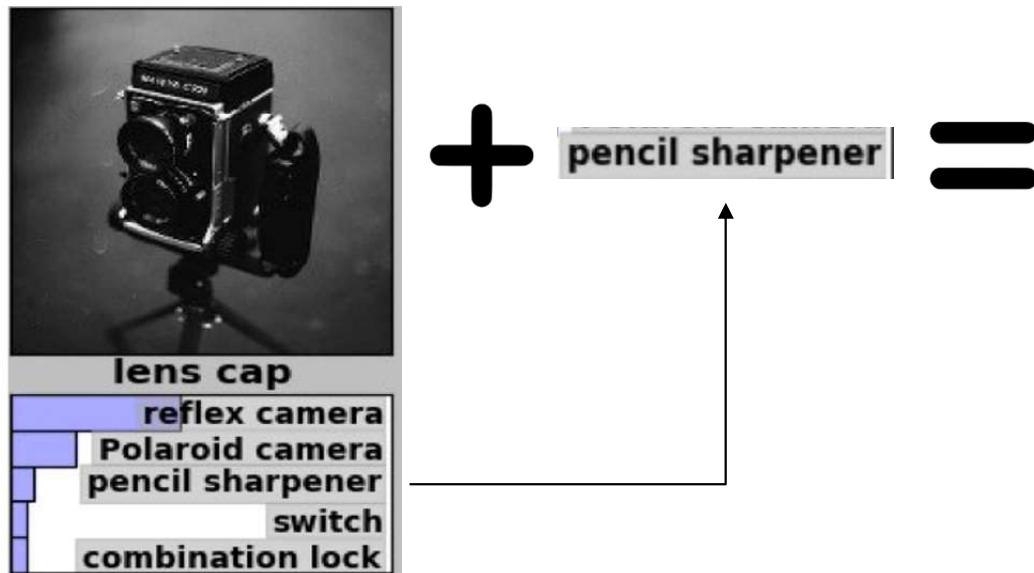
Arcimboldo

Human-made

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). Imagenet classification with deep convolutional neural networks. *Advances in Neural Information Processing Systems*, 25, 1097-1105.

Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2017). ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6), 84-90. doi: 10.1145/3065386

Why might not
these be great
design ideas?



Why might not these be great design ideas?



“...product development with humor that stimulates the curiosity as a theme and design.”

Interior Flaner Shop

★★★★★ 4.49 (5,778)

Shop Information

Contact Shop

Classic cameras! In fact pencil cutting ♪ camera pencil sharpener.

Kikkerland camera pencil sharpener (DTL)

KIKKERLAND



10X POINTS Hurry! 9 days left!
Ends 17/04/2017 15:59 JST
Buy it and earn 230 points!

PayPal OK Details

€ 20.17 (¥ 2,376)
Actual price may vary according to exchange rates.
Payment will be charged in yen.

Estimate Shipping Fees

★★★★★ 4.20 (5) [See Product Reviews](#)

Quantity:

* Please note 1 Non-mail flights

Add to Cart

This page is a simplified version of an original Japanese page.
[See the original page](#)

Kikkerland camera Pencil Sharpener

Classic cameras! In fact pencil cutting ♪ camera pencil sharpener.

Kikkerland

Toy manufacturer founded in New York in 1992. Netherlands-born Jan van der lande started business from one vase was designed by Amsterdam-born friend.

Now also including the Netherlands, works of designers from around the world, especially in "daily life you can enjoy" engaged in product development with humor that stimulates the curiosity as a theme and design.

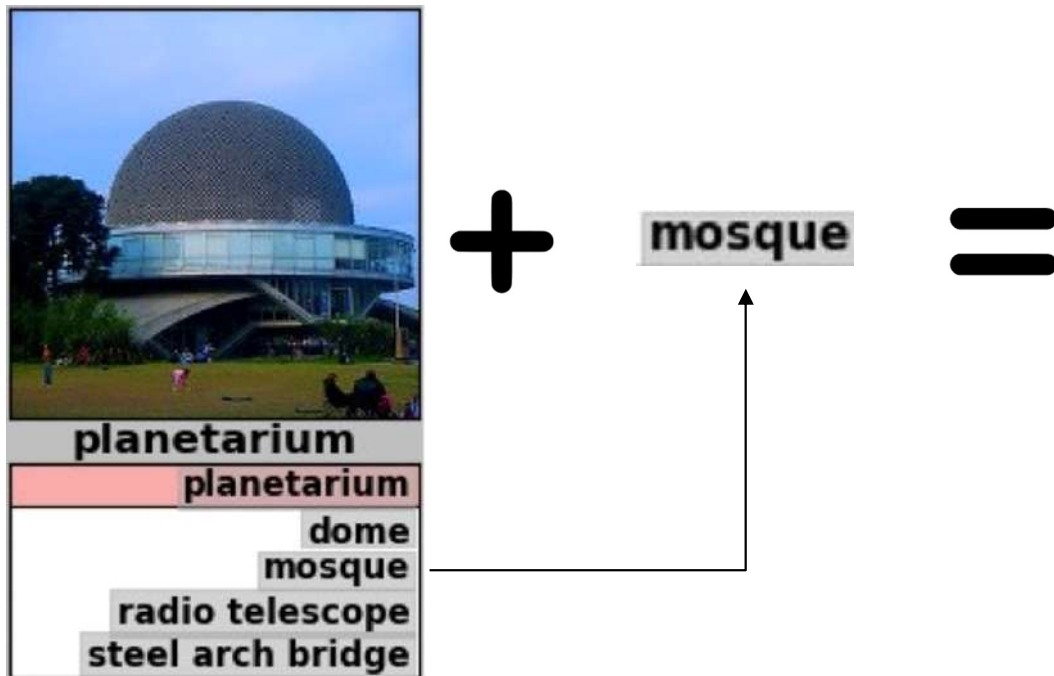
Lots of product introduction, announced Rainbow maker reflects prisms of rainbow-colored room attracting attention around the world

camera
pencil sharpener

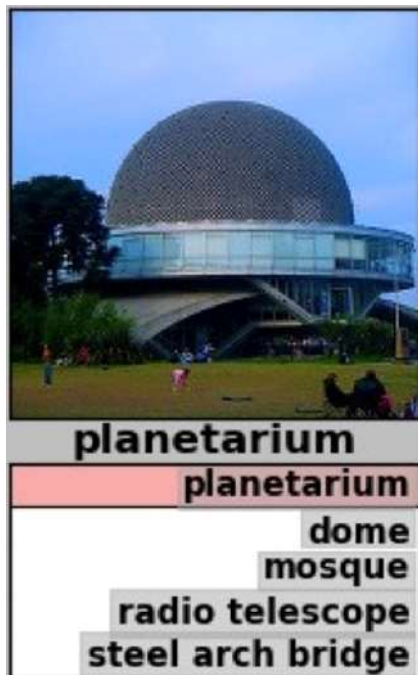


Recommended for those who want to combine relaxing rooms!

Why might not these be great design ideas?



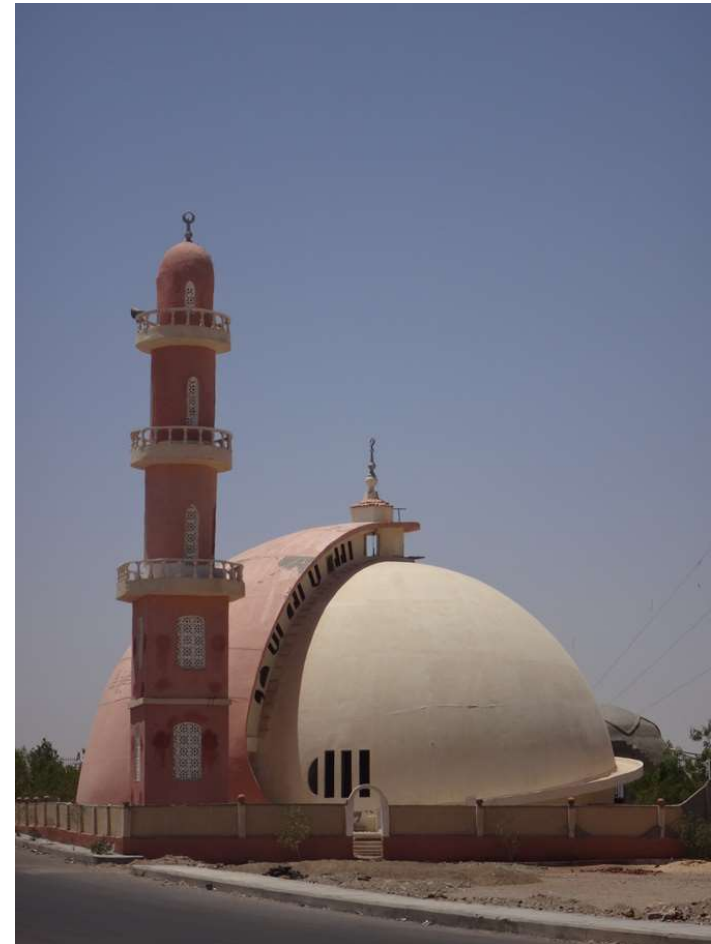
Why might not these be great design ideas?



+

mosque

=



mosque-slash-planetarium



National Planetarium Kuala Lumpur

Location: Kuala Lumpur

Location Info

Address: No. 53, Jalan Perdana, 50480 Kuala Lumpur, Malaysia

Office Number: +603 2273 4301 / +603 2271 4303

Opening Hours: 08:00am - 05:00pm (Monday-Friday)

Location

The National Planetarium is located conveniently inside the [Lake Gardens Kuala Lumpur](#). It was back in 1993 when the planetarium was first opened to the public. The planetarium has been intelligently designed and structured to mimic a mosque with a blue dome. In order to locate it in the large Lake Gardens is easy. The National Planetarium sits in between the Kuala Lumpur Bird Park and the National Museum.

People see a mosque in a planetarium as well...

“The planetarium has been intelligently designed and structured to mimic a mosque with a blue dome.”

... or they see a planetarium in a mosque

11.03.2018

Vladikavkaz Shiites fight Sunnis for ownership of historic mosque – so it becomes a planetarium instead

Zhanna Tarkhanova

Vladikavkaz

JAM
news



RESEARCH-ARTICLE



Computer-vision Classification-algorithms Are Inherently Creative When Error-prone

Author: [Johan F. Hoorn](#) [Authors Info & Claims](#)

VRCAI '22: Proceedings of the 18th ACM SIGGRAPH International Conference on Virtual-Reality Continuum and its Applications in Industry • December 2022 • Article No.: 31 • Pages 1–9 • <https://doi.org/10.1145/3574131.3574444>

Published: 13 January 2023 [Publication History](#)



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ABSTRACT

Whether coming from a linear support vector machine, from logistic regression, or a quasi-Newtonian, the fine-tuning of the decision boundary in any given data set is essential to mitigate the loss term so that neural nets in image recognition can divide a data space into separate sections and correctly classify an input. By their very nature, neural nets are logically non-deterministic but rest on probability-weighted associations, which are adjusted recursively to enhance the similarity of intermediate results to the target output, the remaining difference being the 'error.' However, taxonomies should not be crisp but seen as fuzzy classes, allowing for hybrid exemplars that transgress category boundaries. The associative and similarity orientation of neural nets and deep learning makes such systems inherently creative in that misclassifications are at the basis of creative crossovers in information processing. This new conceptualization of network errors is supported by the ratings of 40 top-ranking designers of 20 image-recognition mistakes on the dimensions of creativity and innovativeness.

Virtual Tutor Summary



Human Information Processing is different from computation because perception, memory, problem solving, and (motor) responses are less 'stable' but also more adaptive

All the visual, textual, sound, and touch input become an integrated whole and recede in memory, sometimes in a single cell (e.g., faces and places)

Memory is not fixed but changes over time (neurons change their connections). Memory is context-sensitive

Problem-solving skills may differ. For deterministic problems, you need reasoning (e.g., means-end analysis). For underdetermined problems, you need creative solutions